Gift bags: environment report

Gym Anything for Robotics · RoboWorld environment `gift-bags`



Two FR3 arms take the lid off the gift box (UNASSISTED scripted joint targets). Frame at 9.4 s of playback, caption bar included.

Environment `gift-bags` (aliases `gift_bags`, `open-gift-bags`) · Newton 1.5 MuJoCo robots + a dedicated implicit contact world for paper, tape, rope and packaging items (`gift_bags/ipcworld.py`, IPC Toolkit 1.6 + MKL PARDISO) · default robot `dual-fr3` (two FR3 arms, Franka Hands), also `dual-panda`, `single-fr3` and `single-panda` (single arms: home preset only) · action group *open gift bags* (16 occurrences, 7 leaf groups).

A robot shop clerk, a robot opening presents and a robot in a stock room. The tasks ask two FR3 arms with Franka Hands (default; a Panda pair and, for the home preset, a single arm are the other configurations) to take a factory-flat laminated rope-handle gift tote off a stack at a boutique counter, open it, square its block bottom and put a board in it; at a dining table, to lift the lid off a set-up gift box (the air drawn through the skirt slot can drag the base up) and to unwrap a taped present; and, at a receiving table, to open a taped supply carton. Paper, creases, tape, rope and the packaging items live in an implicit contact world (shells with orthotropic membranes, creases that remember their history, tearing by vertex splitting, cohesive tape, rods, a compressible paper caliper, a lumped lid air film) stepped beside the MuJoCo robots.

Status. The environment is built and integrated with GAR (CLI, HTTP, dashboard, task resolution; four robot configurations, no humanoid). Of the four activities: lifting the lid is shown completable by the robot, at 1 cm/s only; opening the supply carton is shown only with phantom assistance; unwrapping the present is not shown even with assistance; opening the flat tote is not shown with the holds of two hands.

- *Shown:*
  - friction is Coulomb's law with rest: a block and a sheet of paper stay put below the friction angle and slide above it with the ledger's coefficient to four digits (§3.2);
  - the paper rests at exactly one caliper (§6.1); finger links pressed into paper and boxes with up to 253 N leave the world finite and free of intersections (§10);
  - the lid's air film follows its ODE (descent within 1.5 %) and drags the base up only on a fast lift;
  - the tote's shipping state (flap folded up, pressed, stored a week) is made without a split or a damaged triangle, and the counter world with the rope handles starts from it (§6.4);
  - the lid box, the carton and the tote on its stack come to rest when left alone (the tote moves 0.84 mm in the first ten seconds and 0.002 µm/s after a minute); the wrapped present's paper is still creeping 1.5 µm/s after 30 s, its box 0.03 µm/s (§10);
  - an opened tote stands (§6.7), and released stored energy is never amplified (§6.8);
  - the wrap's out-of-plane tear force is 0.52 N against a target of about 0.45 N (§6.6);
  - packing tape peels at 90° within 2.3 % of the force its own bond energy implies (9.6 N: the ledger's estimate of 200 J/m^2 on a kraft liner times the tape's width; the tape's data sheet gives adhesion to steel only, so this shows the

model books its energy, not that it matches a product), and phantom hands open the taped carton to the task's goal state with the evaluator reporting success, three times with the same forces to seven digits (§6.11);

- the drawn props on the work surfaces are solid: a candle and a hand are stopped by them (§2.2);
- the evaluator passes its positive controls and flips each check on its negative edits (§7);
- **two unassisted FR3 arms take the lid off the gift box at 1 cm/s and set it on the table, and the evaluator reports success**, on one solver thread and on two (§9.3); an unassisted FR3 picks an opened tote (without its handles) up by its rim and sets it down (§9.1).

- *Not shown, or wrong:*
  - **the factory-flat tote has not been opened with the holds of two hands.** One hand lifts it, the other flicks the bottom flap down, the two part the bands: the mouth opens fully and nothing tears. A fist-sized block then goes in and presses, and the bottom does not square: its two ends stay folded in the gusset feet, and sent down into the bottom's fold the block is squeezed out again (§8.2). The openings that do reach the goal state hold 2,760 of the bag's 3,559 vertices: a mould, not hands, and they were made on an earlier start state and friction model. The counter tasks are not shown to be completable;
  - **the present has not been unwrapped.** Phantom hands peel the end tape and open the end; then the paper goes with the box, in nine attempts by sliding and by tearing (§8.3). The wrap does not clamp the box (§6.5);
  - **at 10 cm/s the robot does not get the lid off: it slips out of the squeeze in five runs of five** (§9.3);
  - the robot's opening of one taped end of the present is not shown on the current models (§9.4);
  - the end tapes of the present hold 6-7.5 N at rest in the task world, three times what its generator left on them (§6.9, §13);
  - a 180° tape peel on the carton costs four times the real force, and the office tape on gift wrap peels at 1.2x its energy balance with force spikes (§6.9, §6.11);
  - with paper in contact the simulation runs far above the 200x limit (§11);
  - a joint target sent again with exactly the least duration its refusal names trips the arm's velocity reflex (179 times in one abuse run, §10): the rule has no margin;
  - no robot has opened the tote, unwrapped the present or opened the carton; the Unitree G1 only stands, falls when it reaches (recorded), and is refused as a robot choice (§12).

Every number below was measured in the simulation by the experiment named next to it, unless it carries a provenance label from the parameter ledger (§5).

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1 The real job

Scope: what people actually do when they open gift bags and presents, and which of those jobs this environment reproduces.

The action group gathers 16 recorded activities. By paid-work value, opening bags at a sales counter (Retail Salespersons, "Bag or package purchases and wrap gifts", $1.54 B of the group's $3.37 B) and opening supply cartons in receiving (46 %) dominate; by everyday-life hours, opening presents dominates: tearing the wrap off (483 M h/yr) and lifting a box lid (483 M h/yr, ATUS 120101). This environment reproduces the sales-counter bag, the two present-opening activities and the receiving carton (flaps on score lines, packing tape).

Preset	Setting	Item	Tasks
`counter`	boutique cash-wrap counter, 0.915 m	five factory-flat laminated Euro totes 203 x 102 x 254 mm with rope handles (top one a full paper model in its shipping state, bottom flap folded up; four below as rigid plates), three 200 x 99 mm bottom boards, a boxed candle (420 g)	`open-bag`, `bag-purchase`
`home`	dining table, 0.75 m	a two-piece set-up gift box with a telescoping lid holding a photo frame (seed 0: 180 x 130 x 55 mm, 0.5 mm clearance, 30 mm skirt, 350 g frame; other seeds draw one of three boxes, three lid clearances and two wrap stocks, §7); for the present tasks the box is wrapped in gift wrap with three strips of Magic tape	`lift-lid`, `unwrap-gift`, `open-present`
`receiving`	stainless receiving table in a back room	a regular slotted carton 14 x 10 x 8 in of C-flute board, its top seam sealed with 48 mm 3M 371 packing tape, holding eight tea cartons and a glass jar	`open-carton`

How each is done by people: a clerk takes a flat tote from under the counter, pinches the two faces at the mouth and pulls them apart, pushes a hand inside to square the block bottom, drops in the board and stands the bag up. A present's lid comes off with both hands on the lid; lifted fast, the air drawn in through the fraction-of-a-millimetre slot between skirt and base drags the base up with it. (The air film is measured for a lid sinking onto its base: De Ruiter et al., Phys. Rev. Fluids 7, 044101, 2022, which notes only that the pressure inside is below atmospheric while a lid is lifted and reports no lifted base. The base coming up with a pulled lid is described in patent US5361562: operators "grasp the lid and wait for the body to slip downward".) Wrapped presents are opened by tearing the paper from a seam or by peeling the tape and unfolding it.

The policy sees the instruction verbatim (`task.py:INSTRUCTIONS`), e.g. `open-bag`: "You are at the cash-wrap counter of a boutique. Take the top gift bag from the stack of flat bags, open it, square its block bottom, put one of the bottom boards inside it flat on the bottom, and leave the bag standing open on the dark rubber counter mat. Do not tear the bag. Release everything when you are done."

2 Scene at real scale

Scope: what is in each workcell. Geometry a robot or packaging can touch is physical (`scene.py`, collision boxes in both simulators, items and sheets in the contact world); everything else is visual only (`dressing.py`).

2.1 Physical geometry

Item	Geometry and mass	Contact	Source
Robots	two FR3 + Franka Hand on pedestals at (±0.30, 0.55) m, base at the work-surface height, facing the work area	MuJoCo collision hulls; the same hulls as driven obstacles in the contact world	catalog `gift-fr3`
Counter	1.52 x 0.60 m laminate top at 0.915 m; 400 x 300 x 3 mm rubber mat	both simulators	[A]
Top tote	203 x 102 x 254 mm net, 3,219 vertices (10.2 mm mesh graded to 0.8 mm rows at the bottom folds), 157 g/m^2 coated paper + 18 µm BOPP, 40 mm turn-top with 900 g/m^2 board, glued double bottom; in its shipping state (§6.4), with two rope handles on four knots (∼68 g with them)	contact world shell + rods + rigid knots	`bagnet.py`, `handles.py`, [V]/[D]

Item	Geometry and mass	Contact	Source
Lower totes	four rigid plates with the folded flat tote's 203 x 254 mm outline, 2.5 mm, 69 g each, free bodies; no handles on them, so the stack is 10 mm high where real totes with their handle lumps would stack higher	contact world rigid	[A] disclosed: only the top bag is meant to be lifted
Bottom boards	three 200 x 99 x 0.5 mm chipboard, 7 g	contact world rigid	[V]/[D]
Boxed candle	90 x 90 x 110 mm SBS carton, 420 g	contact world rigid	[E]
Dining table	1.50 x 0.90 m oak top at 0.75 m	both simulators	[A]
Gift box base / lid	1.5 mm greyboard trays, 180 x 130 x 55 mm base (63 g) and 184 x 134 x 30 mm lid (46 g, outer: base + 0.5 mm clearance + 1.5 mm wall per side), 0.5 mm clearance per side	contact world rigid (five-box shells) + lumped air film	`giftbox.py`, [V]/[D]
Photo frame	170 x 120 x 15 mm, 350 g	contact world rigid	[E]
Gift wrap	81 g/m^2 gloss, 1,683 vertices (8 mm), girth wrap with 40 mm seam under the box, ends folded side-flaps / top / bottom (Hallmark order)	contact world shell	`wrapmap.py`, [V]/[G]
Tape	three strips of Scotch Magic 810, 19 mm, on the seam and each end flap	shells bonded by cohesive laws	[V]
Receiving table	stainless work table	both simulators	[A]
Supply carton	RSC 14 x 10 x 8 in (356 x 254 x 203 mm inside), C-flute board; body and four top flaps rigid, the flaps on creased hinges; 48 mm 3M 371 tape over the top seam with legs down both ends	contact world rigid + hinges + tape shell	`carton.py`, [S]/ [V]/[E]
Carton contents	eight tea cartons 150 x 80 x 70 mm, 120 g each; a glass jar 85 x 110 mm, 450 g	contact world rigid	[E]

2.2 Dressing: what is drawn, and which of it is solid

Room shells, the cash-wrap cabinet, merchandise shelving with vases and books, a plant, the shop sign (authored artwork), ribbon spools, tissue, card reader and receipt printer (counter); oak table legs, leather dining chairs, sofa, console with vase and picture frame, pendant lamp, a wall picture and two other wrapped presents (home); wire shelving with cartons, crates, a hand truck and a waste bin (receiving). Models and scanned textures are CC0 Poly Haven assets pinned by SHA-256 in `visual_assets.json` (`python -m gift_bags.visual_assets`). Print artwork (the boutique tote, the wrap pattern, the candle carton, the sign) is generated by `artwork.py`.

Everything drawn on a work surface or within an arm's reach has a collision box in both simulators (`dressing.physical_props`, placed from the same constants as the pictures; `scene.add_props`), so an arm, a carried bag or a pushed board cannot pass through what the overhead camera shows.

Preset	Solid props (static boxes)
counter	ribbon rack, tissue stack, card terminal, receipt printer, tablet stand (on the counter); the cabinet under the counter top; the two display shelf units behind the robots
home	the two other presents on the table; the four table legs; the three dining chairs
receiving	the four table legs and the lower shelf; the two steel shelf units

They are boxes (a chair is the box around it, 0.43-0.58 m wide and 0.97 m high; a shelf unit is solid, with nothing to reach into), and they are fixed: a pushed terminal does not slide and a bumped chair does not move. The paper and the items collide with them in the contact world and the robots in the robot simulator; that each stops something was checked by pushing (the boxed candle dragged against the card terminal, and the robot's hand lowered onto the receipt printer with a free-space control beside it). Not solid: the room shells and floor dressing out of reach (sofa, console, plant, crates, hand truck, waste bin, lamps), and the robot stands, which the arms cannot reach.

At home the arms stand on ballasted stands clad in the table's veneer. The office tape is drawn as a matte film (it shows as a pale strip on the dark glossy paper), and the packing tape is tan. The rope holes are one mesh cell, 10 mm squares, where a real hole is about 5 mm round (§13).

2.3 Start states as rendered

Rendered from the start states after the 0.3 s start settle (RTX path tracing, 48 samples).



Figure 1. Counter preset, room camera: the cash-wrap counter with the two FR3 arms on pedestals behind it, the rubber mat, the tote stack (red, right of the mat), the bottom boards and the candle carton (left).



Figure 2. Counter preset, bag camera: the top tote as it ships. The bottom flap is folded up over the front face (near end), the front rope handle lies on the bag through its two holes, and the back handle runs underneath. The edge of the next tote in the stack shows below it.

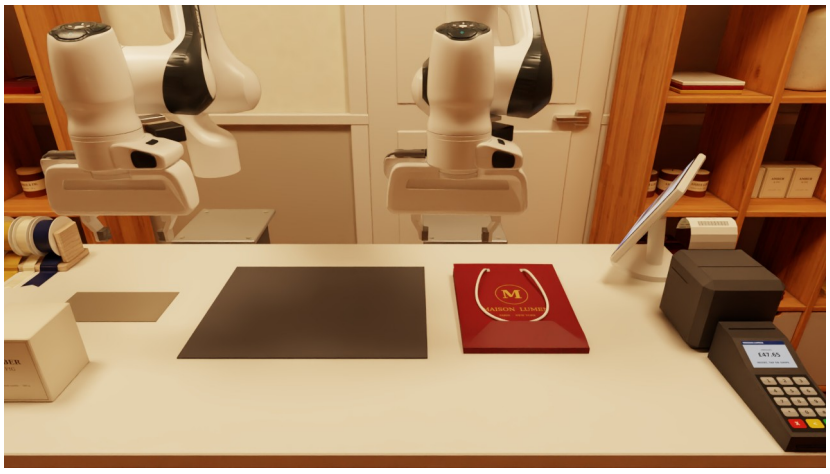


Figure 3. Counter preset, counter_wide camera.



Figure 4. Home preset, lift-lid: the set-up gift box on the dining table; two other wrapped presents are dressing.

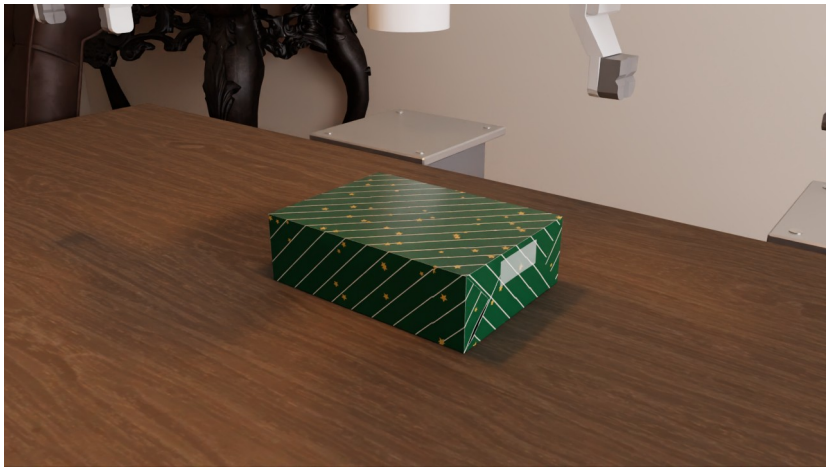


Figure 5. Home preset, unwrap-gift: the wrapped present (81 g/m^2 wrap; Magic tape on the end flap) in its stored start state.



Figure 6. Receiving preset, carton camera: the sealed supply carton on the receiving table, 48 mm packing tape over the top seam with a leg down each end.



Figure 7. Receiving preset, room camera.



Figure 8. Policy cameras, counter preset: overhead_rgb (ceiling mount).



Figure 9. Policy cameras, counter preset: right wrist camera.

3 Paper and packaging physics

Scope: what the contact world simulates and with which equations. Implementation `ipcworld.py`*; parameters* `params.py` *(§5).*

The installed engines have no paper: Newton's VBD cloth stretched a sheet at paper stiffness by 1.3–15 % where paper allows $\sim 10^{-5}$ (measured). The packaging therefore lives in an implicit contact world that is stepped beside the robot simulator:

- **Unknowns:** sheet vertex positions (3 per vertex) and 12 affine coordinates per rigid item (affine body dynamics, orthogonality energy $\kappa_{\text{ortho}} = 10^9 \cdot$volume) and per robot link.
- **Step:** implicit Euler on the incremental potential $E(q) = \tfrac{1}{2} |q - \tilde{q}|^2_M / \Delta t^2 + E_{\text{shell}} + E_{\text{crease}} + E_{\text{bond}} + E_{\text{film}} + E_{\text{drag}} + E_{\text{links}} + \kappa \Sigma w_k b(d_k) + \Sigma$ friction, minimised by Newton's method (Pardiso on the PSD-projected Hessian), CCD-limited backtracking line search, $\Delta t = 4$ ms.
- **Shell:** orthotropic St. Venant–Kirchhoff membrane per triangle (MD/CD stiffness, shear by the Baum rule), discrete hinges for bending (stiffness from $D_{\text{MD}}/D_{\text{CD}}$), panel plastic hinges.
- **Creases:** factory scores and folds are elasto-plastic hinges with two flow mechanisms in series with the elastic stiffness: $m = k(\theta - \theta_1 - \theta_2)$, a fast sinh flow $d\theta_1/dt = \omega_1 \sinh(m/m_1)$ and a slower recoverable one with its own back-stiffness, $d\theta_2/dt = \omega_2 \sinh((m - k_2\theta_2)/m_2)$. The law is fitted to the one-loop folding tests of Nagasawa, Kaneko & Adachi (2019) on 0.30 mm creased board (§6.10: release angle after 0–20 s holds and the creep recovery after release, RMS 1.2 measurement standard deviations against 5.6 for a one-mechanism law) and scaled to each caliper by t^2 (Lechenault, Thiria & Adda-Bedia 2014, measured on Mylar). Thin papers (the wrap, kraft) take the time behaviour of folded printing paper (Farain 2020: the smallest fold angle reached, 36°, and 5.7 per decade of time; what those two numbers do and do not show is in §6.10); the laminated tote keeps the board law. Springback, ageing and relaxation follow from the law, and every hinge's plastic state is part of the saved state.
- **Tearing:** a crack advances by splitting the mesh vertex at its tip when the energy release rate there (virtual crack closure, per mode) reaches the stock's tear energy (in-plane Γ_{I}, out-of-plane Γ_{III}), or the membrane force reaches the strength X; the new vertex pair is held by a cohesive bond that releases Γ per torn length. A tear starts at a free edge on a combined strength and out-of-plane criterion (an estimate). §6.6.
- **Tape:** each tape triangle is bonded to the paper under it at 16 quadrature points (the centroids of its 4 x 4 subdivision) by a bilinear cohesive law (stiffness $1.5 \cdot 10^{10}$ Pa/m, strength 0.40 MPa = the paper's z-direction tensile strength, toughness $\Gamma = 150$ J/m^2, the paper-surface delamination value; damage from 27 µm to 0.75 mm of opening), irreversible damage. Bonds form only where the tape touches the paper in the start state, each with its own separation as rest offset, so every bond starts unloaded (§6.9).
- **Rope handles:** discrete elastic rods (5 mm braided PP rope, 12.5 g/m, EA 15 kN, EI $2 \cdot 10^{-5}$ N·m^2, 400 mm loops) through holes punched in the turn-top band; each end is welded to a rigid knot (14 x 14 x 5 mm) that sits inside the band and cannot pass the hole, so a pull on a handle reaches the bag as contact bearing around the hole.
- **Receiving carton:** rigid C-flute panels; four flaps on hinges that carry the two-mechanism crease law; a 3M 371 seam tape as a shell bonded to the board by a cohesive law (§6.11).
- **Lid air film:** a lumped, quasi-steady laminar slot ($C = \Sigma w c^3/(12 \mu L)$) plus an entrance/exit loss ($K = 1.5$) between lid skirt and base wall, added as an implicit dissipation; equal and opposite on lid and base.
- **Air drag:** quadratic normal drag on free sheet regions (C_D 1.17).

3.1 Contact: a compressible paper caliper

Every contact (paper–paper, paper–rigid, rigid–rigid, robot–paper) is an IPC log barrier with lagged smooth Coulomb friction (ipctk 1.6, area-weighted). Its distances have a physical meaning (`ipcworld.contact_caliper`):

- The caliper t splits into an incompressible fibre core, φ = sheet density / cellulose density (1,500 kg/m^3): $\varphi = 0.80$ for the laminated tote, 0.64 for the wrap. Two midsurfaces never come closer than φt (the barrier's hard distance d_{min}); the barrier acts over the compressible remainder ($\hat{d} = (1 - \varphi)t$). Rigid surfaces are deflated by the core half, $\varphi t/2$.

- The barrier stiffness κ is calibrated so a 1 MPa press compresses the caliper by 5 % [E]: $\kappa = 1.8{\cdot}10^{18}$ (tote), $1.6{\cdot}10^{19}$ (wrap) in ipctk's squared-distance units. The resulting law compresses paper 0.5 % at 10 kPa, 5 % at 1 MPa, 13–15 % at 10 MPa.
- Consequences, measured (§6.1): an unloaded stack sits at exactly one caliper per layer; a sheet rests 58.0 µm (wrap) above a rigid surface as predicted; a robot squeeze compresses paper by micrometres instead of collapsing the gap.

3.2 Friction: Coulomb, with rest

Friction is ipctk's smooth Coulomb potential with lagged normal forces: in each step a contact pair's friction force is its coefficient times the normal force the pair carried at the start of the step, opposing the pair's tangential displacement u, and smoothed below ε = 1 mm/s x 4 ms = 4 µm. Two additions make this Coulomb's law. The friction-slide bench below checks them.

- **Rest (a stick displacement).** As shipped, u is measured from the start of every step, so a pair can carry a tangential load only by slipping again in every step: anything resting on a slope creeps at 1 mm/s x load / (2 x limit). Each collision vertex here keeps the displacement it has accumulated while moving slower than the transition velocity (at most ε long; it fades between 0.1 and 1 x the transition velocity), and u is the step's displacement plus that. A loaded contact below its limit holds with an elastic offset of at most 4 µm and no creep; the potential, its limit and its stiffness are unchanged.
- **Sliding (when the Newton iteration may stop).** A stopping rule on the size of the update alone (a step tolerance, 10 µm in the task worlds) does not let a contact slide. Friction goes from zero to its limit within 4 µm, and at these loads the barrier's normal force doubles within a fraction of a micrometre, so a small update says nothing about either. From rest the first update is load / (limit / ε): under 10 µm for any load below 2.5 x the limit. With that rule it is accepted, the next step starts from zero again, and the contact carries whatever is asked of it while creeping at no more than 10 µm per step: **a block with friction 0.35 stands on a 60° incline, creeping 2.5 mm/s** (the diagnostic row below). With a tenfold tighter tolerance it slides, but against 0.38 to 1.0 instead of 0.35: the normal force left by a small update is always too high (a Newton step on a convex force law lands on the stiff side), and the next step's friction uses it. The iteration therefore ends only if its update also leaves every collision vertex's contact force (barrier and friction, predicted with the contact Hessian the iteration already has) within 5 % of the largest contact force on any vertex; otherwise the update is taken and the iteration goes on. The 5 % is calibrated on the bench below [C]: what it bounds is the last update, and the error left after it is of second order. The effective coefficient is 0.3500 at 1 % and at 5 %, 0.3500-0.3508 at 10 %, up to 2 % high at 20 %, and at 30 % the block no longer starts to slide at 20°.

Block 80 x 60 x 40 mm on a plane, friction 0.35 (friction angle 19.29°), task solver settings	**0.25 kg**	**2 kg**
Plane at 0.9°, 5°, 10°, 15°, 18°, 19°: rate over the last 0.5 s of 2 s	0.0000 µm/s at every angle	the same
... travel before it stopped (it is set down from 0.1 mm, on the slope)	2.6, 15, 52, 114, 369, 1,614 µm	the same
Plane at 20°, 21°, 25°, 35°, 45°, 60°: effective coefficient, at 0.05 to 4.8 m/s	0.3500-0.3502 at every angle and speed	the same
... Newton iterations per step, resting / sliding	1.02-1.42 / 5.3-10.1	1.02-1.42 / 5.3-10.3
DIAGNOSTIC without the stick displacement, 0.9°: creep	22.7 µm/s (formula 22.4)	–
DIAGNOSTIC with a step-tolerance stopping rule only, 20° to 60°: speed reached	0.3-2.5 mm/s (it does not slide)	–

A flat sheet of gift wrap (81 g/m^2, 120 x 96 mm, 8 mm cells) on the plane, friction 0.40 (friction angle 21.8°)	
Plane at 5°, 15°, 20°: rate over the last 0.5 s of 2 s; travel before it stopped	0.0000 µm/s; 14, 74, 302 µm
Plane at 24°, 30°, 45°: effective coefficient, at 0.05 to 2.2 m/s	0.4000-0.4001 at every angle and speed
... Newton iterations per step, resting / sliding	1.03-1.09 / 3.0

What this does not cover: the coefficients themselves are estimates (§5); the 5 % is relative to the largest contact force in the world, so a light contact beside a hard squeeze is converged less tightly than it would be alone; and a step that reaches the iteration cap ends unconverged, with its contact forces not settled (§11 gives how often).

Sliding costs iterations: 5 to 10 per step for this block.

4 Coupling with the robots

Scope: how MuJoCo robots and the contact world exchange motion and force (`world.py`, `rig.py`).

- The robots run in Newton 1.5's MuJoCo solver (CPU, implicitfast, elliptic cones) at 1 kHz; four robot ticks make one 4 ms contact step. Robot–furniture contact is in MuJoCo with solref time constant 0.005 s (§6.3).
- Every robot link's collision hull is a **driven obstacle** in the contact world, pulled to its MuJoCo pose by a spring (10^5 N/m translation, link compliance; $3{\cdot}10^5$ N/m for finger pads) that is force-limited at 250 N (Huber).
- Contact wrenches on the driven links (barrier + friction) are returned to the MuJoCo bodies lagged one step, first-order filtered ($\tau = 5$ ms) and capped at 250 N. Along each hand's closing axis (taken from the finger's own frame) the equal-and-opposite part of the two fingers' forces is the grasp squeeze, internal to the hand; their net axial force is external (a closed hand pushing an item with an outer finger face) and is applied once, at the pushing fingers, to the body that carries the hand (the FR3's wrist link). Inside the four robot substeps of a contact step the returned force is extrapolated to first order along its push direction, $F = F_0 + k\,(p_0 - p)\cdot n$ with the coupling-spring stiffness k, never pulling and capped (`world.py _predict_contact_wrench`). With this coupling a bimanual lid squeeze holds a steady ∼20 N through a lift (the squeeze bench, §9.3).
- **Franka Hand:** position mode (half-gap targets 0–40 mm, 50 mm/s per finger) or force mode (closes at 50 mm/s until blocked, then holds the commanded force, ≤ 35 N per finger). The closing coordinate is resolved in the contact world against whatever is between the fingers; the MuJoCo fingers follow it.
- FR3 joint-velocity reflex: a joint faster than 1.05x its limit freezes its arm's targets.

5 Parameter ledger

Provenance labels: V vendor, S standard, T handbook, M measured (peer-reviewed), D derived, E engineering estimate, A authored, C calibrated here, F fitted here to published data by a named probe, U a seller's listing of the product in use (weaker than a manufacturer's data sheet), W read from a product web page.

Quantity	Value	Label	Source / reasoning
Tote laminate	173.5 g/m^2, 145 µm; E_t 0.89 / 0.44 MN/m (MD/CD); D 1.23 / 0.56 mN·m; X 7.9 / 3.9 kN/m	M (proxy stock)/V/D	157 g/m^2 coated paper with moduli from Nayakanti, Tawfick & Hart 2018 (E_1 6.8, E_2 3.1, G_{12} 2.17 GPa), measured on Daler-Rowney Canford 150 g/m^2, an uncoated card: a proxy stock. A denser coated art paper with a BOPP skin is expected to be stiffer in plane and in bending [E, direction only], so the model is likely on the soft side; + 18 µm BOPP (Arrsys)
Turn-top band	+ 900 g/m^2 chipboard: E_t 6.1 / 3.2 MN/m, D 0.80 / 0.43 N·m	V/E	Onlineprinters CLASSIC rope-handle bag, onlineprinters.ie/p/paper-bags-with-rope-handles-classic-10x36x10cm (retrieved 2026-10-06): "a stabilising 900 gsm chipboard reinforcement in the turn top", "190 gsm coated paper", "glued bottom board made of 350 gsm chipboard". This is a heavy band; retail turn-tops with 425 g/m^2 board also exist. Board moduli E
Gift wrap	81 g/m^2, 85 µm; E_t 0.43 / 0.17 MN/m; X 4.0 / 2.0 kN/m; tear energy Γ_{III} 1.0 J/m (out of plane), Γ_I 0.30 J/m (in plane)	V/E/D	Uline S-8551 basis weight (50 lb commercial counter roll); moduli and strength E; Γ_{III} = 2 x Elmendorf force, 0.3–0.6 N [E]; Γ_I from fracture-toughness index 3.4–4.0 J·m/kg (Mäkelä, coated and newsprint papers) x grammage [D]
Consumer gift wrap (home variation)	53 g/m^2, 58 µm; X 2.6 / 1.3 kN/m; Γ_{III} 0.65 J/m, Γ_I 0.20 J/m	V/D	Holmen GIFT Silk 53 g/m^2 technical data (bulk 1.1 cm^3/g); strengths and tear energies scaled by grammage from the 81 g/m^2 stock [D]
Tote tear energies	laminate Γ_{III} 1.6 J/m, Γ_I 1.0 J/m; turn-top band 15 / 6 J/m; glued bottom 2x the laminate	E	paper ply 2 x Elmendorf from tear index 3.5–5.0 mN·m^2/g x 157 g/m^2, bracket 1.1–2.0 J/m; no vendor tear figure found for the laminate

Quantity	Value	Label	Source / reasoning
Rope handles	5 mm braided PP rope, 12.5 g/m, EA 15 kN, EI $2{\cdot}10^{-5}$ N·m^2, loop 400 mm; knots 14 x 14 x 5 mm, 1.2 g; holes 40 mm from the centre line, 25 mm below the rim	D/E/A	PP rope 0.91 g/cm^3 at ∼70 % fill (supplier coils 11.0–11.4 g/m for 6 mm [V]); stiffnesses E; loop and hole positions A. A hole is one 10.2 mm mesh cell (a real hole ∼5 mm)
Magic tape	65 g/m^2, 56 µm, E_t 0.1 MN/m	V/E	3M Scotch Magic Tape 810 product sheet, the edition hosted at static.rapidonline.com/pdf/12-9149_ v1.pdf: 56 µm, 2.41 N/cm on steel (ASTM D3330), tensile 22.8 N/cm, 35 % elongation (another edition lists 60 µm, 2.5 N/cm, 26 N/cm, 25 %); acetate modulus E
Tape on wrap	k $1.5{\cdot}10^{10}$ Pa/m, σ 0.40 MPa, Γ 150 J/m^2	E / V range / E	k: adhesive layer (label ledger); σ: paper z-direction tensile strength, 150 g/m^2 kraftliner 300–550 kPa (ISO 15754 / TAPPI T541, ABB L&W ZD tester documentation); Γ: min(adhesive 480 J/m^2, paper-surface delamination ∼150 J/m^2)
Creases (two-mechanism law, §6.10)	at 0.30 mm board: k 0.864 N·m/m/rad, m_1 14.6 mN·m/m, ω_1 $4.27{\cdot}10^{-7}$ rad/s, k_2 0.271 N·m/m/rad, m_2 8.64 mN·m/m, ω_2 $3.23{\cdot}10^{-3}$ rad/s; tote (laminate, board law) k 0.202, m_1 3.40 mN·m/m; wrap (paper law) k 0.069, m_1 1.06 mN·m/m, ω_2 $2.60{\cdot}10^{-3}$ rad/s	M fitted (C) / M scaled (E)	fitted to Nagasawa et al. 2019 Figs. 12–14; moments ∝ t^2 (Lechenault 2014, M, proxy material Mylar); paper time factors fitted to Farain 2020 (printing paper; a weak anchor, §6.10)
Friction (per material pair, `params.FRICTION`)	tote–tote 0.35, tote–counter 0.40, tote–mat 0.60, wrap–wrap 0.35, wrap–tape 0.40, wrap–box cover 0.35, box–box 0.35, box–oak 0.40, board–counter 0.40, Franka fingertip–paper/board/items 0.50, robot link covers 0.30, PP rope–paper/board 0.30, rope–mat 0.50, rope–rope 0.25, ASSISTED phantoms 0.30	M range / E	paper on paper 0.2–0.9 across paper samples and measuring conditions together (de Silveira & Hutchings 1997; the method alone moves a value by 15–30 %, and no coated or laminated paper was in that study, so the laminated tote's and the gloss wrap's values have no measured range behind them); values inside such ranges E; the Franka Hand manual does not state the fingertip material (hard ridged plastic: 0.5, range 0.3–0.6, E). The contact world uses exactly this table (`friction_strict`: an unlisted pair is refused; a test asserts the built table equals the ledger).
Contact caliper	$\varphi = \rho/1500$ (0.80 tote, 0.64 wrap, 0.71 box cover); 5 % compression at 1 MPa	D/E	§3.1
Lid film	air μ $1.81{\cdot}10^{-5}$ Pa·s, ρ 1.204 kg/m^3, slot loss K 1.5	T/E	handbook; laminar slot Re ∼250
Gift box	greyboard 1.5 mm; base 63 g, lid 46 g; clearance 0.5 mm per side (seed 0; home seeds draw 0.5 / 0.8 / 1.0 mm)	V/D/E	clearance sensitivity §6.2
Home variation boxes	the authored 180 x 130 x 55 mm set-up box with its 0.35 kg frame; a Uline 7 1/2 x 7 1/2 x 2 in window gift box with an IKEA FISKBO 10x15 frame (0.11 kg); a Uline 8 1/2 x 11 x 2 in set-up box (S-10708, .032 in board) with a FISKBO 13x18 frame (0.15 kg)	V/E	Uline product pages (catalogue sizes read as inside dimensions [E]; the window box's board caliper 1.5 mm [E]); IKEA package weights taken as the frames' masses [V]; the authored frame's 0.35 kg is E
Robot coupling	link 10^5 N/m, pad $3{\cdot}10^5$ N/m, cap 250 N, reaction filter 5 ms	E	§4; pads after the label environment
MuJoCo robot–furniture contact	solref time constant 0.005 s	C	≤ 2 mm transient link penetration (§6.3)
Contact step / Newton tolerance	4 ms / 10 µm per step	A/C	§11

6 Calibration and component validation

Scope: benches that check one mechanism each against a prediction or a measured value.

6.1 Contact caliper (drop test)

An 81 g/m^2 wrap sheet is dropped on a table and a 300 g box is put on it. Predicted by the caliper (§3.1): the sheet midsurface rests 58.0 µm above the table and the box bottom 116 µm above it (one sheet-to-rigid rest distance on each side of the sheet). Measured: 0.058 mm and 25.116 mm − 25.000 mm = 0.116 mm, one Newton iteration per step at rest, kinetic energy $2.6{\cdot}10^{-15}$ J.

6.2 Lid air film

- **Self-weight descent** (rigid only): the lid released with 1 mm of skirt engaged sinks onto the base as air leaves through the slot. Simulated vs the same lumped slot law integrated as a scalar ODE: 0.334 / 0.329 s (1.0 mm clearance), 1.526 / 1.534 s (0.5 mm), 6.43 / 6.47 s (0.3 mm): ratios 0.993–1.015. This checks the implementation of the film inside the implicit step, not the physics of the film itself; vendor descriptions of slow-sinking lids are qualitative.
- **Fast and slow lift** (ASSISTED phantom hand, §8.1): 10 cm/s produces 193 Pa of suction and 4.5 N on the base, more than base + frame weigh (4.05 N), and the base lifts 1.3 mm; 0.5 cm/s produces 9 Pa / 0.22 N and the base does not move.
- **Clearance decides the outcome** (ASSISTED phantom lift of the lid by 60 mm, the frame inside, judged by the task's limits: no point of the base's bottom more than 5 mm up, at most 10° tilt). The 0.5 mm per side is an estimate (makers describe "fractions of a millimetre"; no measured set-up-box tolerance was found) and the film's suction scales with the cube of the gap:

Clearance per side	1 cm/s: peak suction / force on the base	base rise	10 cm/s: peak suction / force	base rise
0.3 mm	84 Pa / 1.96 N	0	285 Pa / 6.67 N	**43.9 mm** (carried up with the lid; task failed)
0.5 mm (scene)	19 Pa / 0.44 N	0	200 Pa / 4.67 N	1.2 mm
0.8 mm	6 Pa / 0.15 N	0	68 Pa / 1.58 N	0
1.0 mm	4 Pa / 0.09 N	0	42 Pa / 0.98 N	0

The model's gap is uniform along the 30 mm skirt. A real lid's gap tapers and the flow is controlled by its narrowest part (De Ruiter et al. 2022); a uniform gap equal to the narrowest one over-predicts the suction, so for a given narrowest gap the model errs on the strict side. The taper itself is not modelled.

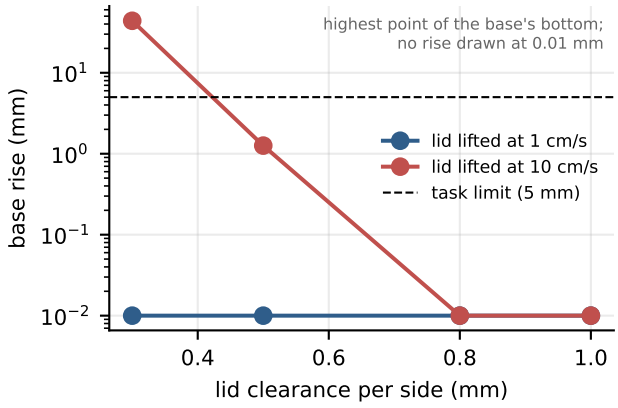


Figure 10. Highest point of the base's bottom against the lid clearance, lid lifted at 1 and 10 cm/s (zero rise drawn at 0.01 mm).

6.3 Robot squeeze and MuJoCo contact stiffness

- MuJoCo alone: an FR3 wrist driven into a wooden block sinks 6.45 mm (worst transient) with solref time constant 0.02 s and 1.85 mm with 0.005 s; both settle to 0 mm. The environment uses 0.005 s.

- Coupled world: the same joint path sweeps the wrist across gift wrap lying on the block, a legal action a policy can take. The path's end point is 2.6 mm above the block: the wrist grazes the sheet. The sheet slides 1.3 mm out of the way and stays there; the peak link reaction is 1.7 N, the smallest gap 28.7 µm against 31.0 µm at rest, 11 of 500 steps at the iteration cap, finite, no intersections. What a hard squeeze does is in the abuse runs (§10): finger links pressed into the wrapped present, the lid box and the carton with up to 253 N (the coupling's limit), finite and without intersections.

6.4 The tote's start state: as it ships, with its handles

A Euro tote ships flat with its bottom flap folded up once more against a face, and has sat that way for days. (Gussets tucked in and the bottom folded against a panel is how flat-bottom gusseted bags are packed: US8486500B2 describes it, for a woven polypropylene bag. No source was found that shows it for a laminated rope-handle paper tote; it is an estimate [E].) That history is the start state: it sets what the creases remember when the clerk opens the bag.

The mesh (`bagnet.bag_mesh(graded=True)`): rows beside the bottom perimeter folds start at 0.8 mm and grow by 1.6x to the nominal 10.2 mm cell, the same sequence along every bottom fold, so the 45° bottom and gusset diagonals still run through mesh nodes; 3,219 vertices, the flat-state map isometric, one 10.2 mm cell left open for each of the four handle holes.

How the state is made (`bagnet.shipping_state`; no fixture folds the bag):

1. *Authored fold.* The folded geometry is written down (`fold_section`): each of the four stacked layers turns the fold on its own nose, the noses a similar family so that neighbouring layers' legs are parallel 0.95 of a layer pitch apart; the layers start 1.5 pitches apart at the fold. This geometry is not isometric within 12 mm of the fold (edges up to 3x their length); it only has to be free of contact violations.
2. *Relaxation.* On a flat table under the storage pressure, the bag's rest shape is blended from that geometry to the true paper over 0.5 s (the procedure that makes the wrapped present, §6.5), then the fold settles 0.2 s, so the layers slide and find their own radii under contact. **Nothing can tear or be damaged during the blend and the settle**: the strains of a half-blended rest shape are not strains of paper.
3. *Crumpling.* Where the fold crosses the tucked gussets, joined layers turn it on different radii, so an inner layer has more length than its path (π t per layer step, 1.4 mm between the innermost and outermost of four); and an inner layer's first mesh row, a straight chord, is longer than the arc and flat its paper takes inside the next layer's nose. Real paper takes such surplus as wrinkles and curvature in the inner layers. The 0.8 mm mesh cannot draw either, so while the bag is made, paper compressed in its plane beyond its yield strain (0.5 % [E]: paper crushes at about half its tensile strength) hides the excess as **slack**: the triangle keeps its true length and is stress-free anywhere between its crushed and its true length (`ipcworld.slack_response`). A triangle that has begun to wrinkle goes on at a tenth of that load [E] and hides at most half its length [E]. There is no yield in tension: paper pulled beyond its strength tears, and a state with a tear is refused.
4. *Press.* As press rolls set a fold, the pressure rises to 200 kPa and returns to the storage pressure (0.6 s). The 200 kPa is an estimate [E] with one argument behind it, an internal one: an inner layer's surplus stands as an arch about 4 mm high over 30 mm, and an arch carrying the paper's crushing load stands up to about 160 kPa, so only a press above that flattens the fold. It is not taken from a press.
5. *Storage.* Held at 650 Pa (about 50 totes above it in a carton [A]) for a week [A]: every crease starts with the memory of the bag as it was formed (its open shape) and is aged at its folded angle by the crease law; panel bending and in-plane stress relax by paper's stress relaxation (0.05 per e-fold of time, 67 % in a week; a panel hinge at its plastic limit sets fully, as a crease does).
6. *On the counter.* The pressure goes; the bag lies free for an hour [A], integrated in growing time steps, so the creases' recovery after release is over before the task starts.

In a task the slack behaves like wrinkled paper (`IPCWorld._slack_post`): pulled, a crumpled triangle opens without force up to its true length and is paper beyond it; wrinkles pulled out do not come back by themselves (the slack shrinks to what is left); pushed, it wrinkles again only beyond a tenth of the yield strain. Paper without slack has no in-plane plasticity in a task. A strip drawn at 90 % of its length with that slack: held at 90 % and 95 % it carries 2-6 N/m, at 100 % 1.7 N/m, at 100.2 % 460 N/m; released and pushed back it buckles like any strip (a single check of the law, not kept as a bench).

Result (built from scratch, no saved state):

Quantity	Value
vertex splits (tears) / fully damaged triangles	**0 / 0**
triangles with slack	967 of 6,336; 659 within 12 mm of the fold line, 405 at its two ends. Most are slight: median crushed length 97.8 % of the true length; 184 below 90 %, 21 below 70 %, 2 at the limit of 50 %
largest hidden length in one triangle	1.4 mm
largest elastic stretch	0.94 % after the settle (before the press), 0.50 % at the end of the hold, 0.13 % at rest on the counter; no triangle above its strength strain (0.87 %) at the end of the hold
thickness at the flap while stored at 650 Pa	3.5 mm
flap tip after release (1 s) / after the hour on the counter	14.8 mm / **15.7 mm** above the table
flap tip speed at the end	−0.007 mm/s
fold-line crease: angle / plastic set / recoverable part	101° / 102° / 3.7°
steps at the iteration cap	72 of 1,175 (during the blend, the settle and the press)
build time, from scratch	2,178 s (36 min) on four solver threads at a load average of 20-45; the handle assembly after it 743 s

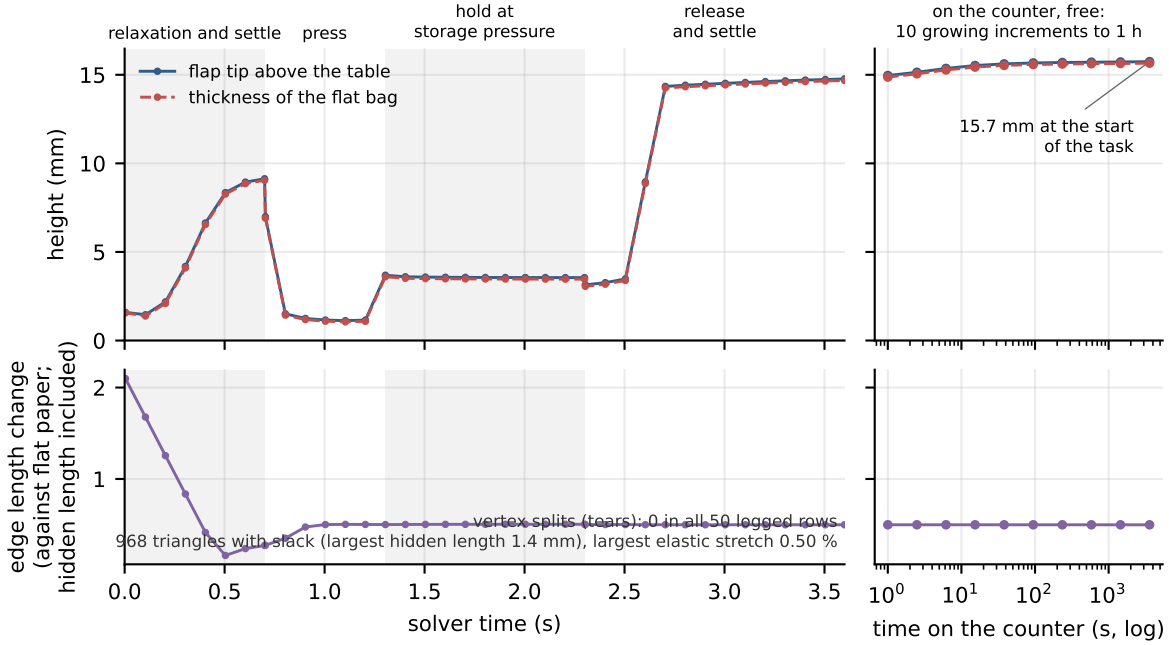


Figure 11. The tote's shipping state being made: height of the folded flap's top and the bag's thickness through the relaxation, the press, the stored hold, the release and the hour on the counter.

Reproducibility. Two builds of the state from scratch on the same code differ by 15 µm at the median vertex and 1.35 mm at most: the parallel solves are not bit-reproducible and crumpling amplifies the difference. The scenes use a stored build (`python -m gift_bags.prebuild --build` makes one).

The flap does not lie flat when nothing presses on it: it rests 15.7 mm up. What lifts it is the part of the fold that sits in the rows beside the score line (the fold-line crease itself carries 102° of the 180°) and the creases' recoverable set. Whether a real tote's free flap rests at 5 mm or at 16 mm was not measured; in a stack or a carton it lies flat.

Rope handles (`handles.py`): two 400 mm loops of 5 mm PP rope (rods) and four knots (rigid, 14 x 14 x 5 mm) inside the turn-top band. Each rope end is a node at its knot's centre, inside the bag; the rope's first segment crosses the paper through the hole (one 10.2 mm cell left open), and the rest of the loop lies outside. The end nodes are part of the knot (welded to it, no contact between a rope's end and its own knot), and a knot cannot pass its hole, so a pull on a handle is carried by the rope bearing on the hole's rim and the knot bearing on the

paper around it. The rope has its own contact compliance (0.5 mm). The handles are assembled into the flat bag by blending its rest shape from an authored geometry (bag lifted, front face bulged over the knots, loops laid flat) to the true paper while everything settles on the stack (0.3 s blend, 3.0 s settle; with a 0.5 s settle the ropes are still relaxing 30 s into a task): the flat bag is 19.8 mm thick at its knots, the lump a real flat tote has at its handles; 50 of the 825 assembly steps ran at the iteration cap, all during the blend; at the end of the settle the fastest rope node moves 10.6 µm/s, the fastest knot 5.6 µm/s and 0.05°/s, the fastest paper vertex 36 µm/s (the folded flap, whose creases are still recovering); built in 743 s (`start_state.handles` in the scene geometry; file `flat_tote_handles_*.npz`).

The attachment is tested by loading it (ASSISTED phantom hook on the five middle nodes of the front loop):

- *Lift.* The hook rises 0.50 m in 3 s and holds 1 s. The whole tote leaves the stack and hangs from the handle, its lowest point 82 mm above the stack; every knot stays within 1.8 mm of its hole, measured in the paper's plane; nothing tears; the hook carries 0.66 N. The hanging mass is 68 g (0.67 N): bag 52.7 g, ropes 10.6 g, knots 4.8 g.
- *Pull.* With every paper vertex then held where it is (0.5 N cap each), the hook's force is ramped to 40 N: the handle holds, knots within 1.9 mm of their holes, no tear (392x wall/sim, 86 of 2,075 steps at the iteration cap). Its pull-out force is above 40 N and was not found. (The record's hook force is the sum of the five node forces, which during the lift also counts the parts with which the nodes pull against each other, 2.0 N; in the pull all five are at their caps and point the same way.)

A lift of only 0.15 m drags the bag along the stack: the loop is slack until its middle is 190 mm from the holes.

The start state is the whole settled stack. The four lower totes are free plates authored 50 µm apart. While the handles are assembled they settle onto each other (20.7 µm per gap, the top plate 82.8 µm lower), and the bag and both rope loops settle with them. The assembly returns the plates' settled poses and the scene uses them, so the ropes start outside their contact minimum. The task world as built: closest paper pair 132.8 µm (minimum 115.7), closest rope pair 2,430 µm (minimum 2,087 with the rope's own compliance); 25 steps at rest take one Newton iteration each (the 10 s rest is in §10).

6.5 Wrapped gift start state

The wrap map folds the sheet onto the box analytically (Hallmark order) with layers one pitch apart; it is not isometric where four layers meet at the box ends (6.6 % peak stretch). Start state = that geometry relaxed with the tapes bonded and normal friction while the rest shape morphs to the true flat paper (0.2 s + 0.08 s hold, 70 steps): peak stretch 0.63 % afterwards (below the 0.94 %/1.18 % damage onset), every tape bond intact; 32 of the 70 morph steps hit the iteration cap. The tapes are bonded and friction is on during the relaxation because an untaped, frictionless relaxation lets the end folds fall open.

Storage (`wrapmap.relaxed_gift`). The present was wrapped a day before it is opened [A]:

- *Paper.* Held in its wrapped shape, the wrap's in-plane stress and its panels' bending relax by paper's stress relaxation (0.05 per e-fold of time: 57 % in a day), and its creases age by their law, as the tote's do in its shipping state (§6.4). The largest in-plane set is 0.36 %. The present re-equilibrates for 0.2 s.
- *Adhesive.* A pressure-sensitive bond under a steady load creeps. Twelve times, the bonds are laid again where the tape now lies, unloaded, and the present settles 0.1 s [A]. The most loaded tape's bond force falls 43 N (the generator's bonds) → 10.1 → 6.8 → 5.4 → … → 2.1 N; it is still falling by 6 % per cycle at the end. The task world receives the stored history (the wrap's hinge state and in-plane set) and lays its bonds on that state (§6.9). Without the storage each end tape holds 15-18 N at rest (§6.9).

The paper as cut (`wrapmap.girth_allowance`). The fold map's net is the bare box's net: its girth is the box's, 381.1 mm. But the paper's middle surface lies one sheet-on-rigid rest distance (63 µm) off each face, so its path around the four long edges is 8 x 63 µm = 0.504 mm longer. Cut to the bare box, the sleeve is 0.132 % short; the day of storage relaxes 57 % of that, leaving 0.057 % (243 N/m at the wrap's 425 kN/m). Measured in the stored state such a sleeve carries 0.025-0.047 % (117-223 N/m): about 190 N of clamp on the box, and about 67 N to slide it. The paper's true girth therefore includes that path plus 1 mm [E]: a wrapper lays the seam by hand, to about a millimetre, and cannot tape under tension. The tension is zero for any slack of zero or more; the 1 mm only sets how far the paper stands off.

Wrapped present, authored box, 81 g/m² wrap (`start_state` in the scene geometry)	**paper cut to the bare box**	**paper cut with the allowance**
girth tension in the sleeve, middle 70 % of the box, per face	117-223 N/m	at most 2.8 N/m, mean 0.28 N/m
paper's largest distance from the box there	the rest distance	1.6 mm
most loaded tape's net bond force at rest, in the generator (box without its photo frame)	2.2 N	2.3 N
most loaded tape's bond tension in the task world, frame in the box (§6.9)	2.3 N	**7.5 N** (net 5.4 N)
every tape bond intact	yes	yes
build	613 s	896 s (on a throttled CPU quota, §11)

Over 16 of the 18 variants a home seed can draw (two boxes, three lid clearances, two wrap stocks; §12 says which are shipped), the largest girth tension is 1.1-3.0 N/m, the paper stands 0.7-1.6 mm off the box at most, and the most loaded tape holds 0.3-2.4 N. Each start state records its own values (`girth_tension_n_per_m_max`, `standoff_mm_max`, `bond_force_after_adhesive_n`).

6.6 Tearing

A crack band on the tensile membrane strain gives a trouser-tear force of 4.8–6.9 N for the wrap whatever its tear energy (Γ 0.25 / 1 / 4 J/m: 6.2 / 5.9 / 6.9 N; twice the strength: 15.2 N): the force follows the in-plane strength, which is the wrong driver for an out-of-plane tear. Tearing is therefore modelled by **vertex splitting** (`ipcworld.py`):

- A crack advances by splitting the mesh vertex at its tip. At each tip the energy release rate is measured per mode from the nodal force and the opening (virtual crack closure, $G = P{\cdot}D/2\ell$) and compared with the stock's tear energy, mixed between the in-plane Γ_I and the out-of-plane Γ_{III} by the mode share; a tip also splits when the membrane force across it reaches the strength X.
- The new pair of vertices is held by a cohesive bond that releases exactly $\Gamma{\cdot}\ell$ as it opens, so a torn length ℓ costs $\Gamma{\cdot}\ell$ whatever the mesh.
- A tear starts at a free edge when $\sqrt{((\bar{N}/X)^2 + (P_{III}/(1.05{\cdot}\Gamma_{III}))^2)} \geq 1$. $\bar{N}$ is the membrane force across the prospective crack line averaged over the stock's own length $l_c = (2/\pi){\cdot}E_t{\cdot}\Gamma_I/X^2$ (the average-stress length of Whitney and Nuismer, derived from the ledger: 8.1 / 5.6 mm for the wrap along MD / CD, 18.5 / 10.0 mm for the tote laminate, 84 / 64 mm for the turn-top band); P_{III} is the out-of-plane pull between the two sides, compared with the traction at which a running tear's bond is inserted on the trouser bench (1.02–1.12 Γ_{III}). That an uncut edge starts at that traction is an estimate; no out-of-plane edge-initiation measurement was found.

Propagation (trouser tear, energy balance $F = \Gamma_{III}/2$):

Bench	**Result**	**Target**
wrap trouser, 8 mm (the scene mesh)	0.519 N; 0.954 J/m booked	0.50 N (Γ_{III} 1.0 J/m); ∼0.45 N Elmendorf-class [E]
wrap trouser, Γ 0.5 / 1 / 2 J/m	0.263 / 0.519 / 0.986 N	the force follows Γ
tote laminate trouser	0.927 N	0.80 N (Γ_{III} 1.6 J/m [E])
tensile strip (ISO 1924 geometry)	0.92–1.05 X	X
notched tension (SENT), onset / peak	0.92–0.93 / 1.08–1.11 Γ_I	Γ_I
at rest, 1 s (wrapped present, flat tote)	0 splits	0

These benches check that the implementation books Γ and X correctly; Γ_{III} itself comes from an estimated Elmendorf range (§5), so they do not validate the material.

Mesh dependence (48 mm strip, 72 mm of tear):

Cell	**Force, bending yield as in the scene (plastic work at the legs)**	**Force with elastic legs**
8 mm	0.524 N (0.009 J/m)	0.521 N
4 mm	0.512 N (0.142 J/m)	0.476 N
2.5 mm	**0.998 N** (0.366 J/m; 41 of 55 splits off the centre line)	0.561 N, all on the line

With elastic legs the tear holds 0.95–1.12 $\Gamma/2$ from 8 to 2.5 mm with a straight crack: the splitting is mesh-independent. The drift comes from the panel hinges' yield law: once the cell is smaller than the legs' yield radius ($t/2\varepsilon_y = 8.5$ mm) the legs' fold localises into one hinge row, its plastic work per crack length grows roughly as 1/cell, and the creased legs turn the loading at the tip in-plane, so the crack wanders and branches. That is a mesh-objectivity defect of the perfectly plastic hinge. **Accepted band: ±3 % for 4–8 mm cells. Below 4 mm the tear force is not qualified** (2.0x at 2.5 mm), so a tear that crosses the tote's 0.8 mm fold rows is not qualified. The force trace is a sawtooth: at 8 mm, 4 of 280 steady steps exceed 2 N (peaks to 3.5 N, 9 % of the impulse; mean without them 0.48 N), one per element, because the criterion is tested after each 4 ms step.

Initiation (two rigid grips either side of an uncut edge point of the wrap, pulled apart out of plane; first split):

Grip	Edge test	8 mm	4 mm	2.5 mm
fingertip-sized (one edge vertex each side)	averaged over l_c (the model)	2.84 N	2.13 N	2.19 N
fingertip-sized	one element (for comparison)	2.84 N	0.80 N	0.49 N
hand-sized (16 mm deep)	both	6.47 N	5.66 N	5.17 N

If the in-plane part of the test reads one element's membrane force, the starting pull under a concentrated load falls with the cell. Averaged over l_c it does not (2.13 and 2.19 N at 4 and 2.5 mm; the 8 mm value is 30 % higher: coarse elements beside a point load carry a spurious in-plane index). With a hand-sized grip the start is governed by the out-of-plane term and is mesh-stable either way (the 20 % drift from 8 to 2.5 mm includes a 1 mm difference in the grip gap). An edge tear in the wrap therefore starts at 2–3 N under a fingertip and 5–6.5 N under a hand-sized grip, and then runs at 0.5 N. The level of the out-of-plane term is the estimate named above: **initiation is not calibrated against a measurement.** Every bench of this section, run again, reproduces its numbers to the digits given (wrap trouser 0.519 N, tote trouser 0.927 N, tensile strip 0.98 / 1.02 X at 8 mm and 0.98 / 1.06 X at 4 mm, notched onset 0.92 / 0.93 Γ_I or 59.5 and 58.4 N, the area-grip starts, no split at rest), with one exception: the fingertip start on 2.5 mm cells is 2.34 N, against the 2.19 N in the table. That pull is the out-of-plane case, which uses the mean of the two leaves' normals without normalising it (§8.2). The wrapped present has no split at rest and its wrap's largest in-plane strength index is 0.33.

A single-point pull does not tear the tote: one rim vertex of a clamped turn-top band pulled out of plane with 6 N or 20 N gives no split (in-plane index 0.05 / 0.17); the plain laminate at 6 N none (0.13).

6.7 Can an opened tote stand? (world capacity for the counter task)

The world itself is asked whether the goal state exists. The top tote is built in its open manufactured shape (panels planar, gussets flat, block bottom square) standing on the mat, and nothing touches it; the robots hold still. The rope handles are left out of this test. Two crease states bracket a real opening:

Crease memory	Tilt	Bottom flat (limit 85 %)	Mouth ratio (limit 0.80)	Gusset opening, median (limit 120°)	Drift in the window	Failed checks
worked (that of the open shape), 3 s	0.13°	100 %	1.000	179.7° / 179.8°	0.002 mm	`board_in_bag` only (no board in this test)
factory (that of the shipping state: folded flat, stored a week), 10 s	0.11°	94 %	0.861	143.3° / 142.9°	0.006 mm	`board_in_bag` only

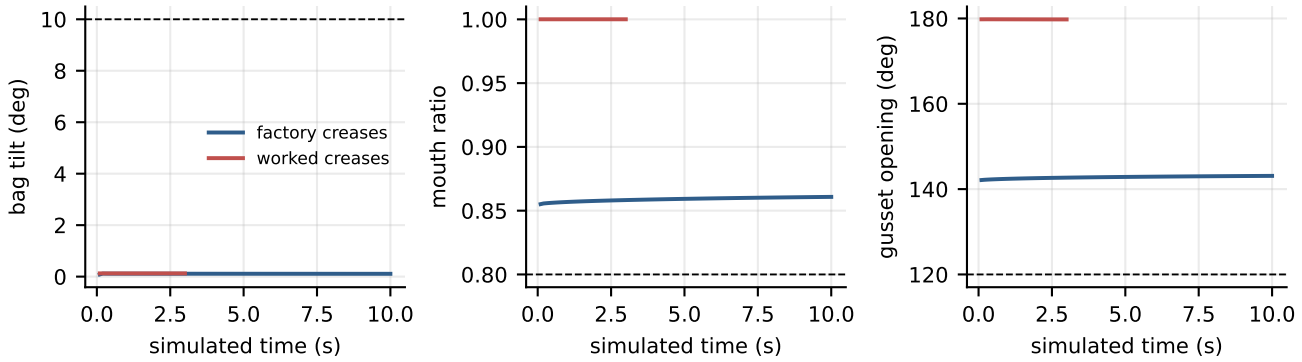


Figure 12. The opened tote left alone on the mat. Dashed: the evaluator's thresholds (tilt 10°, mouth 0.80, gusset 120°).

Both stand, square and at rest: an opened bag stands in this world, and with the shipping state's crease memory it passes every check but the board. The gussets settle near 143° (the crease law's springback from flat) and the base moves less than 0.01 mm in the window. With the paper-class crease law in place of the board law the same test gives 93° (§6.10, a run on an earlier friction model), so the gusset angle depends on an ungrounded choice. The test runs at 35–40x wall/sim.

6.8 Energy audit of a stored-energy release

In assisted attempt 4 (§8.2) the tote leapt ∼13 cm when a phantom frame that had held it down let go. To check that the solver does not create energy, a bench stands the opened factory-crease tote on the mat, presses its rim down 20 mm with the same kind of frame (one spring of 1 kN/m with a 1 N cap on each of the 92 rim vertices), holds 0.5 s and lets go, recording every step the kinetic energy and the potential energy of the contact world without the phantom springs (gravity, shells, creases, contact barrier). After release the total never increased (largest step change -5.4 µJ); the 99 mJ stored by the press was dissipated within 1 s; the bag's peak kinetic energy was 8.3 mJ and its lowest point rose 10.5 mm; the frame pressed with 92 N at most. Attempt 4's leap therefore came from energy stored in a different state (walls buckled over an unsquared bottom flap, 19 mm taller than the open bag), not from the integrator.

The attempt-4 state itself, measured (DIAGNOSTIC: attempt 4's own bottom-extended start without handles, on the current contact world). Every step the work of the phantom springs (their capped force times the vertices' displacement) and of the paddle (its capped drive force times its displacement) is summed; from the hand's withdrawal on, the kinetic and the potential energy of the contact world without the phantom springs are recorded:

Quantity	Value
rim frame force at the release (1 N cap per rim vertex)	69.9 N
work done by the phantom springs / the paddle over the whole run	2.15 J / 0.95 J
potential energy at the release above the bag's final rest (elastic and gravity)	0.431 J
largest change of kinetic + potential energy over one step after the release	−0.38 µJ (it never rose)
energy dissipated in the 1.1 s after the release	0.431 J
peak kinetic energy of the bag after the release	74.6 mJ (17 % of the stored energy, 2.4 % of the fixtures' work)
peak rise of the bag's centroid / of its lowest point	45.2 mm / 34.5 mm

The released bag hops and falls: its energy comes from what the 70 N press stored, and the solver dissipates it. The same replay also tore the tote (15 vertex splits, 162 mm of new crack, `bag_not_torn` false); this run did not record where and when. The contact world records tear events (time, sheet, rest position; `world.extremes()["tear"]`). This press-and-release was not run on the shipping-state tote with handles; the staged opening of §8.2 was run there.

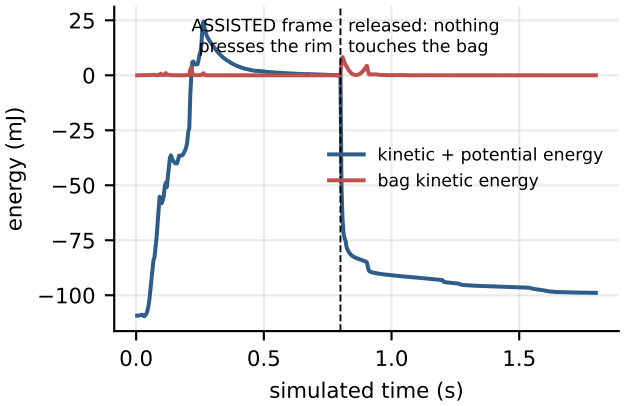


Figure 13. Kinetic plus potential energy of the contact world relative to the moment of release (dashed), and the bag's kinetic energy. Before release the ASSISTED frame does work on the bag; after it the total only falls.

6.9 Tape peel

The bond strength is the paper's z-direction tensile strength (0.40 MPa, §5), so the law is consistent: damage onset $\delta_0 = \sigma/k = 27$ µm with the $1.5{\cdot}10^{10}$ Pa/m adhesive stiffness, full separation $\delta_f = 2\Gamma/\sigma = 0.75$ mm. `CohesiveLaw` refuses any law with $\delta_0 \geq \delta_f$. A strength taken as Γ / 20 µm = 7.5 MPa would put damage onset (0.5 mm) after full separation (40 µm): every bond would stay fully intact up to 0.5 mm of opening (traction up to 7.5 MPa, ∼1.9 kJ/m^2 stored, 12x Γ) and fail outright beyond, and a tape would not peel.

A peel bench peels a 19 mm Magic-tape strip bonded over 50 mm from wrap held flat (ASSISTED clamp, pinch drawn up at 10 mm/s and kept over the peel front, as a 90° peel tester does). The whole bond peeled; the bonds dissipated 0.1425 J, exactly Γ times the bonded area; the mean force over the peel was 5.3 N against the energy-balance prediction $\Gamma w = 2.85$ N (1.8x), delivered as stick-slip: rows of bonds 2.5 mm apart (the tape mesh) let go in snaps with force spikes up to 21 N between stretches near 0.25 N. The peel's process zone is ∼0.6 mm long (bending-dominated, $\ell \approx (8EI\Gamma/\sigma^2)^{1/4}$ with EI $1.3{\cdot}10^{-5}$ N·m, Γ 150 J/m^2, σ 0.40 MPa), a quarter of the tape mesh, so each row's release is a jump.

Quadrature bonds (`tape_bonds.QuadBonds`, `params.TAPE_QUAD = 4`, the default). Each tape triangle is bonded at the centroids of its 4 x 4 subdivision (16 points, each carrying 1/16 of the triangle's area and energy), so the front passes four bond points per mesh row. The same bench:

Bonds	Steady force (energy balance 2.85 N)	Largest spike	Standard deviation	Dissipated
one per tape vertex	5.24 N (1.84x)	22.3 N	7.40 N	0.1425 J
4 x 4 per tape triangle	3.44 N (1.21x)	11.8 N	3.47 N	0.1425 J

The snaps are halved and the mean comes within 21 % of the balance; they are not removed (spikes still 3.4x the mean), because the tape's own mesh (2.5 mm) still sets the shape of the peel arm. Resolving the 0.6 mm zone needs ∼0.5 mm tape cells (about 25x the tape triangles). The alternative, lowering the strength until the zone spans three rows (Turon et al. 2007), needs $\sigma \approx 2$ kPa, 180x below the measured tack, and the tape would no longer hold in shear: rejected. Cost in the wrapped present at rest (run side by side): 0.684 s per contact step with quadrature bonds (16,384 bond points) against 0.556 s per vertex (603).

Bonds as laid (`tape_bonds.bond_tapes(as_laid=True)`, used by the home scene). A pressure-sensitive tape bonds where it touches and starts unloaded: a quadrature point is bonded only if it lies within the law's elastic range of contact, with its own separation as rest offset. Where a tape bridges a flap edge it lies 0.1–0.3 mm above the paper, far beyond the law's 27 µm elastic range. Bonded at the nominal contact offset instead, those bonds start stretched: an instrumented end-flap bench finds 8–12 % of the bonds of all three tapes softening before any load (damage up to 0.68–0.91) and the seam and far-end tapes carrying 46 and 56 N of bond tension at rest.

Stored stress under the end tapes. On a present wrapped without storage history, each end tape holds 19.0-19.1 N after 1 s at rest (the seam tape 4.8 N). A force balance on each tape (contact force by finite rigid shifts of the tape) gives 16.3 N of contact push from the paper under an end tape against 16.3 N of bond pull, and 3.1 N

on the seam tape: the end closure pushes the tape outward and the tape's bonds hold it. Laying the bonds again, unloaded, lets the flap move 0.2-0.35 mm and the load falls from 15 to 4 N in six such cycles: the push is the paper's stored in-plane stress, a stiff spring with a short travel. It comes from the missing storage history and from the generator's own bonds (every tape point within 0.6 mm pulled to the nominal offset: 52 N on the most loaded tape), which lock the layers under the tape in compression. The start state therefore has the history a wrapped present has (§6.5): a day of paper stress relaxation, and the tape's adhesive relaxed. With storage relaxation alone the end tapes hold 10.1 and 11.8 N; the adhesive cycles take that to about 2.2 N. What a real end tape holds was not measured; a closed end can be held shut with a fingertip.

In the task world the end tapes hold three times what the generator leaves on them (a defect of the start-state generator, not corrected):

Tape	Points bonded	Softening / failed	Largest opening	Bond tension after 1 s at rest (net bond force)
seam	91 %	0 / 0	6.4 µm	2.6 N (1.5 N)
−x end	87 %	0 / 0	8.8 µm	6.2 N (4.5 N)
+x end	85 %	0 / 0	16.2 µm	7.5 N (5.4 N)

The generator's own last adhesive cycle left 2.3 N net on the most loaded tape (§6.5), so the load appears when the task world starts. Two checks: it is not the friction memory a new world starts without (tensions change by under 1 % when the stick displacements are set to zero again); **it is the photo frame.** The generator wraps the box empty; the task world puts the 350 g frame in it. With the frame lifted off the base's floor by a phantom grip carrying its 3.43 N, the base rises 0.30 mm and the tensions fall to 1.2 / 2.3 / 2.6 N, the generator's values; set down again they return to 2.6 / 6.4 / 6.1 N. The box sinks 0.3 mm into the loose paper under it (the 1 mm of girth slack) and the end closures take that up. No bond softens, the present rests (§10), and the tape still peels under a 7 N fingernail (§8.3), but every run on the wrapped present starts with its end tapes loaded 6-7.5 N. The remedy is to wrap the box with its contents' weight in `wrapmap.relaxed_gift`; it changes all 18 start states and every record on the present, and was not made (§13).

6.10 Creases: bench and fit against the source

The crease law decides whether an opened tote stands and how a wrapped end springs open, so it is benched against its own source, the one-loop folding test of Nagasawa, Kaneko and Adachi (JAMDSM 13(1), 2019) on 0.30 mm creased coated board (228-237 g/m^2, E_{MD} 5.72 GPa, 15 mm strips, folded to 90° at 0.02-0.4 rps, held 0-20 s, turned back until the load cell reads zero). The data used are their Eqs. 5-7 and their Figs. 12-14, digitised from the PDF at 400 dpi by marker colour (reading error about 0.2° against their 0.7° sample standard deviation).

A single-mechanism law does not match. `m = k(`$\theta - \theta$`_p)`, $\dot{\theta}$`_p = `ω_0 `sinh(m/m_a)` reproduces the 90° moment (0.214 vs 0.215 N·m/m) and its 1 s relaxation (0.176 vs 0.175) by construction. It misses the rest of the data:

- it gives a release angle of 53.4° after a 1 s hold, against the measured 48.5°;
- it has no creep recovery after release, where the measured fold keeps opening 46.0 → 42.7° in 0.42 s and about 5° per decade after that;
- it unloads along a straight line, where the measured unloading is curved.

Its RMS residual over 93 measured points is 5.6 measurement standard deviations.

The law has two inelastic mechanisms (`ipcworld.CreaseLaw`, `crease_return`): a permanent set, plus a recoverable set held by a back spring:

$$m = k_e\,(\theta - \theta_1 - \theta_2), \quad \dot{\theta}_1 = \omega_1 \sinh(m/m_1), \quad \dot{\theta}_2 = \omega_2 \sinh((m - k_2\theta_2)/m_2)$$

The back spring is Nagasawa's own reading of the release: the inner plies bulge while folded and give that energy back on unfolding. Fitted by least squares: k_e 0.864 N·m/m/rad, m_1 14.6 mN·m/m, ω_1 $4.27{\cdot}10^{-7}$ rad/s, k_2 0.271 N·m/m/rad, m_2 8.6 mN·m/m, ω_2 $3.2{\cdot}10^{-3}$ rad/s. The RMS residual is 1.2 standard deviations.

- **Release angle against hold** (0/1/2/3/5/10/20 s): model 45.5-60.6°, measured 46.0-59.4°.
- **Moment at 90° and after a 1 s hold** at 0.02/0.2/0.4 rps: within 0.006-0.017 N·m/m.
- **Not represented:**

- the first-fold transient peak (Mp1 0.244 vs 0.226 N·m/m), which is ply delamination on a fresh crease, while the scene's creases are factory-folded;
- the fresh crease's initial loading slope (about 2 N·m/m/rad in their Fig. 8; the fit gives 0.86);
- the earliest recovery, which runs about 2° fast in the first 0.1 s after release;
- the recovery after a 20 s hold, which is up to 3° slow at 10 s.

The contact-world bench folds a 15 mm strip of the same board in the shell solver with the test's fixture motion and reads the moment from the fixture. It reproduces the 0-D law: release angles 46.5 / 50.3 / 54.3 / 60.3° after 0 / 1 / 5 / 20 s holds, against 46.0 / 48.5 / 54.5 / 59.4° measured. It is mesh-independent: the release angle is 47.1° at 7.5 mm cells and 46.5° at 2.5 mm.

Thin paper and laminate are grounded separately. Scaling every constant by $(t/0.30\ \text{mm})^2$ keeps the angles thickness-independent:

- **For crease stiffness** that is measured on creased Mylar 150-500 µm: $\kappa \propto B/h$, $L^* \approx 200$ h; the rest angle "depends weakly" on h there (fitted 27°, 43° and 39° for the three thicknesses), so thickness-independent angles are an approximation (Lechenault, Thiria & Adda-Bedia, PRL 112:244301, 2014; arXiv 1404.1243) [M, proxy material: polymer].
- **For the fold-opening rate** it is measured on Mylar and paper: independent of thickness and force (Farain, Phys. Rev. Applied 13:014031, 2020) [M].
- **The springback level is not the board's for paper, but the anchor is weak.** Farain gives 36° for printing paper and 50° for Mylar as the smallest fold angles reached at saturating press force. The text does not say how long after release the paper's 36° was read; for Mylar its figure shows about 21° three seconds after folding and about 48° after 48 hours. This report uses the 36° as the angle a paper fold settles at after a 3 s press, which the source does not state. Folds then open logarithmically for days at 5.7 ± 1.5 for paper (the paper prints no unit; read as degrees per decade) and 0.10-0.12 rad per decade for Mylar.
- **The board law on that protocol gives 49° and 0.7°/decade.** So paper holds a pressed fold better and keeps opening longer than the board says.
- **Gift wrap and kraft** therefore use a paper-class law. Its time-behaviour constants are refitted to the two printing-paper numbers, keeping the t^2 moment scale (factors on m_a, k_2, m_2, ω_2 of 0.91 / 0.50 / 2.11 / 0.81 [C]). Two data points do not determine four factors; the solution nearest the board law is used.
- **The BOPP-laminated tote keeps the board law [E], and that choice is not grounded.** One argument for it is that its polymer skins make it Mylar-like (50° measured, 49° from the law). That matches Mylar's angle only. Mylar's opening rate in the same paper is 0.10-0.12; read as radians per decade that is 5.6-7.1° per decade, about the paper value and eight to ten times the board law's 0.7°; read in the units of paper's 5.7 it is about seven times slower than the board law. Neither reading is the board law. And the tote stock is 88 % paper by caliper (127 µm paper, 18 µm film), so the paper-class law is at least as defensible. No fold-recovery data for laminated art paper was found.

Bracket: the tote on the paper-class law instead of the board law (everything else unchanged; both columns run with an earlier friction model, so the board-law column differs from §6.4 and §6.7; the blend phase of the shipping state resumed from the board-law run's, the laws differ from the press on):

	board law (shipped)	**paper-class law**
Shipping state: tears	0	0
flap tip while stored at 650 Pa	4.1 mm	4.5 mm
flap tip after release / after the hour on the counter	18.1 / 18.4 mm	18.8 / 19.0 mm
fold-line crease: angle / set / recoverable	94.3° / 95.7° / 5.7°	93.6° / 95.0° / 5.4°
Opened tote left alone 10 s (factory creases, §6.7): tilt	0.14°	0.63°
mouth ratio (limit 0.80)	0.868	0.803
bottom flat (limit 0.85)	0.947	1.00
gussets' median opening (limit 120°)	140.2° / 141.5°	93.1° / 94.4°, still rising 0.6°/s at 10 s

The folded flap's springback does not depend on the choice (0.6 mm in 18): it comes from the rows beside the score line, not from the crease's time law. **The opened tote does**: with the board law a tote opened once holds its gussets at 140° and passes both gusset checks; with the paper-class law they fall back to 93° and fail them

until the creases are worked, and the mouth sits on its limit. Paper holds a pressed fold better, so a week of flat storage leaves more of the folded memory. The board law stays the shipped law [E]; the task's 120° gusset limit lies between the two outcomes, so whether an opened but unworked tote passes the gusset checks is not established by this environment. Measuring a real tote's gusset angle after one opening would settle it.

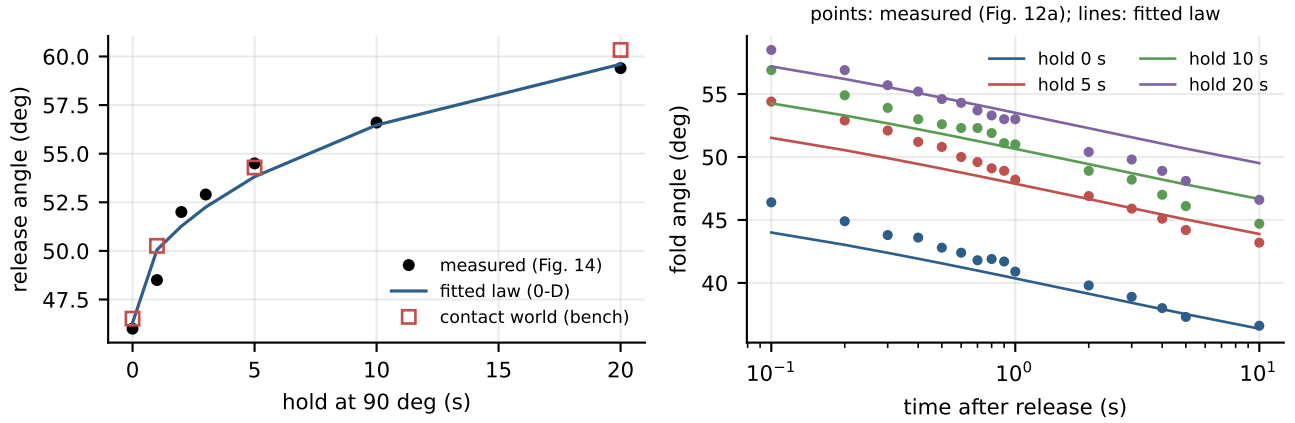


Figure 14. Left, release angle after a 90° fold held 0–20 s: measured (Nagasawa et al. 2019, Fig. 14), the fitted law (0-D) and the contact-world bench; right, the fold reopening after release, measured (Fig. 12a) and the fitted law.

6.11 Receiving carton: tape peel, flap hinges, assisted opening

Model (`carton.py`): rigid C-flute panels; four top flaps on hinges that carry the two-mechanism crease law scaled to a score moment of 0.8 N·m/m [E], aged closed for a day; the 48 mm 3M 371 seam tape a shell bonded to the board by a cohesive law (Γ 200 J/m², the kraft liner's delamination value).

Tape peel (a strip peeled from a board at a prescribed jaw motion). The scene's law regularises the bond strength to the tape mesh ($\sigma = 2\Gamma/h$, derived in `carton.tape_law`): with the ledger's 0.45 MPa the peel force is 71 N against the 9.6 N energy balance.

Peel	Mesh	Steady force	Energy balance $\Gamma w/(1 - \cos\theta)$
90°	2 / 4 / 8 mm (8 mm is the scene's)	9.57 / 9.49 / 9.38 N	9.60 N (0.996 / 0.988 / 0.977)
180°	4 mm	**18.7 N, peaks to 100 N: FAILED**	4.8 N

At 90° the pull work equals the booked dissipation within 0.2 %; the force is a sawtooth from 0 to ∼19 N per mesh row. At 180° the lumped front row is loaded in shear and lets go in bursts: **the tape law is valid for peeling at about 90° only**. Bond damage is updated within the step: frozen within a step, it gives 12.96 N at 90°.

Flap hinge (a major flap, aged closed, opened by a phantom pinch on its free edge and released; the carton body clamped, since an empty 0.33 kg carton left free slides 121–167 mm and tips 23° under the 2–3.5 N pull).

Opened to	After 0.1 s	After 4 s	≥ 90° over the last 3 s
(untaped, no pull)	0.6°	1.2°	no
90°	37.5°	34.1°	no
120°	67.4°	64.0°	no
150°	98.5°	95.0°	yes (5° margin)
180°	129.6°	126.1°	yes (36° margin)

A released flap springs back about 55° at every angle, the crease law's springback of a fresh fold; the contact world agrees with the 0-D law within 0.04°. The flap's weight moment (0.045 N·m per metre of score) is small against the crease's 0.83 N·m/m, so the crease, not gravity, sets where a flap rests: a flap folded right back stays past 90°, one only stood upright falls back. The estimated score moment does not decide this: at 0.3 / 0.8 / 2.0 N·m/m a flap opened to 180° settles at 128.5 / 126.1 / 124.5° and one opened to 150° at 94.0 / 95.0 / 94.5°; only the opening

force scales with it (0.9 / 2.4 / 6.0 N). A minor flap cannot be opened first: its free edge runs under the closed majors (0.4° at the 30 N cap).

Assisted opening to the task's goal (labelled phantoms, robots idle, the task's own evaluator). A phantom hand presses the carton with 20 N; a phantom pinch takes the tape's courtesy tab and peels the whole seam at 90° (one end leg, the top seam, the other leg; 7.7–10.5 N mean, peaks to 25 N); the majors, then the minors, are folded out to 180° in pairs (2.4 N and 1.7 N); 3.4 s hands off. **Evaluator: success, all nine checks**: tape stations joining the majors 47 → 0 (bonded area 201.7 → 0 cm^2, dissipation 4.03 J = Γ times the area); lowest flap angles in the window 125.5° (majors) and 126.4° (minors) against 90°; carton moved 0.97 mm (limit 30); contents moved 3.6 mm (limit 10), none outside. 15.1 s simulated at 128x wall/sim; 98 of 3,862 steps at the iteration cap, all while the loose tape is swung or laid down (another run, at 174x under a heavier host load, gave the same flap angles).

Negative control (same peel, flaps opened only to 120°): the evaluator reports failure (score 0.78). The two major flaps fall back to 70.9° (lowest in the window 70.6°, limit 90°) and fail their checks. The two minor flaps, pushed past the half-open majors (9.4 N instead of 1.7 N), come to rest leaning on them at 99.2°, so their two checks read open although those flaps would not stay open alone. The task still fails: a minor can only lean on a major that is itself below 90°. But the score counts the two propped minors as passed, so it overstates partial progress; **the evaluator does not test whether a flap is propped up by another flap.**

One of twelve starts of the assisted opening locked. After its 0.2 s of rest, from the first step in which the pinch pulled, the collision check reported a pair at 1.8 µm against its 33 µm minimum separation and allowed no motion; the run went on for 200 steps (to 1.00 s) with the pinch pulling 63 N on tape that could not move, 197 failed projections of the contact matrix, and no bond released. The other eleven starts of the same command ran normally. **The cause is not known**: a pair got 65 µm inside its minimum distance in one accepted step, which the barrier should make impossible, so it depends on something that differs between runs of one command (the parallel solves are not bit-reproducible, §9.3). A step that starts with a pair inside the collision check's minimum separation stops the world with a fault that says so, instead of stepping in place.

Not shown: a robot doing any of it; opening by slitting the tape (the common act in a stock room): only peeling from the courtesy tab is modelled, at about 90°; the jar's loose kraft wrap.

7 Task evaluation

Scope: how `task.py` *judges an episode. Every judgment is computed from simulator state; thresholds are authored [A].*

Every task first requires a **hands-off window**: the last 3.0 s of the episode with no robot link touching any packaging item (`checks.hands_off_window`). "At rest" criteria are judged over that window. For bag-purchase the explicit evaluation completes that window itself, counts only contact with the job's items and reads "at rest" over its last 2 s (below).

Task	Check	Physical quantity	Threshold
open-bag, bag-purchase	`mouth_open`	area of the convex hull of the rim vertices projected on the plane normal to the bag axis, over W·G; bag-purchase also: the least front-to-back gap of the rim across its middle 160 mm (pinch guard)	$\geq$ 0.80; gap $\geq$ G/2 (50.8 mm)
	`standing`	bag axis (bottom centroid → rim centroid) from vertical; bag-purchase: signed (an upside-down bag reads 180°)	$\leq$ 10°
	`bottom_squared`	bottom-panel area within 3 mm of the mat plane, and the panel normal from vertical	$\geq$ 85 %, $\leq$ 10°
	`on_mat`	bottom centroid inside the 400 x 300 mm mat	inside
	`gusset_left/right_deployed`	gusset centre-crease opening (180° flat, 0° folded) over its length; open-bag only (bag-purchase reports the fractions and medians, not judged)	$\geq$ 120° on $\geq$ 80 %
	`base_at_rest`	drift of the bottom centroid over the window	$\leq$ 2 mm
	`board_in_bag`	sampled board underside within 4 mm of the bottom panel's plane and over it; bag-purchase: and inside, i.e. at each counted sample the bottom panel's paper lies beneath the board's mid-plane and none of it above	$\geq$ 90 %

Task	Check	Physical quantity	Threshold
open-bag	`stack_undisturbed`	largest move of the four lower totes (bag-purchase: reported as `stack_max_move_mm`, not judged)	$\leq$ 20 mm
open-bag, bag-purchase	`bag_not_torn`	new crack length in the tote since the episode start: each split edge counted by the fraction of its bond's fracture energy that has been dissipated (a split whose bond never opened counts nothing), plus fully torn triangles	$\leq$ 5 mm
bag-purchase	`candle_in_bag`, `candle_upright`	all eight carton corners inside the bag's convex hull; signed tilt of the carton's print-up axis (+z) from vertical	all; $\leq 15°$
	`candle_on_board`	the carton's lowest corner above the top face of a board that passes `board_in_bag`, its centre over that board	0-5 mm (-1 mm allowed)
lift-lid	`lid_off`, `lid_on_table`	lid rim above the base rim (overlap ≤ 0) or clear of the base footprint; lid lowest point $\leq$ 5 mm above the table	
	`base_not_lifted`, `base_not_tipped`	whole-episode peak rise of the highest point of the base's bottom face (any end of it, so a base raised at one end fails); peak tilt	$\leq$ 5 mm; $\leq 10°$
	`frame_in_base`	photo frame centre inside the base	
unwrap-gift	`unwrapped`	fraction of the box's outer surface with paper (intact triangles) within 3 mm	$\leq$ 5 %
	`box_closed`, `box_upright`	lid overlap; base tilt	$\geq$ 20 mm; $\leq 15°$
open-present	`unwrapped`, `lid_off`, `lid_on_table`, `frame_out_on_table`	as above; frame outside the base, lowest point $\leq$ 5 mm above the table	
	`base_upright_on_table`	where the base ends: its tilt, and its lowest point above the table. The present may be picked up and turned over on the way, and the instruction says so (lift-lid's whole-episode rule, base never more than 5 mm up, would forbid the usual way of unwrapping).	$\leq 10°$; $\leq$ 5 mm
open-carton	`seam_open`	seam length still joined by tape bonded to both major flaps. The instruction says how: there is no knife, the tape is peeled from its folded-back tab (a slit is not modelled, §13)	$\leq$ 10 %
	`flap_major_yneg_open`, `flap_major_ypos_open`, `flap_minor_xneg_open`, `flap_minor_xpos_open`	flap angle from closed, through the whole hands-off window	$\geq 90°$
	`carton_in_place`, `carton_upright`	carton body displacement from the start; tilt	$\leq$ 30 mm; $\leq 10°$
	`contents_in_place`	every content item inside the carton and moved	$\leq$ 10 mm

Success = every check true; the reported score is the fraction of true checks (not a success verdict), except for bag-purchase (below).

bag-purchase (it shares `_eval_bag` with open-bag).

- *Instruction:* "You are at the cash-wrap counter of a boutique, bagging a customer's boxed candle. Take the top gift bag from the stack, open it, square its bottom and put one of the bottom boards in it. Leave the bag standing open on the dark rubber counter mat with the boxed candle inside, upright on the board. Release everything when you are done." It states the end state without a sequence and does not say "Do not tear the bag" (ordinary workmanship); `bag_not_torn` still judges it.
- *Checks:* the board must lie on the bottom panel's inner side, so a board on the mat under the bag fails `board_in_bag`. `candle_on_board` reads that the carton rests on a board, so a carton hovering 130 mm up, or wedged on a waist of the gussets, fails. The upright checks are signed: a carton on its lid and a bag upside down fail. `stack_undisturbed`, a 20 mm limit nobody is told and the job does not imply, is reported, not judged.
- *Open is judged one way, by the mouth:* `mouth_open` (hull ≥ 0.80) plus a pinch guard, the least front-to-back face gap across the middle 160 mm at every height of the faces ($\geq$ G/2). Without the guard a mouth pinched to 5 mm at its middle reads hull 1.0, and a body pinched to 9 mm 60 mm below an open rim has an open rim. Recorded opened totes are narrowest at the rim (76-102 mm). The gusset checks are reported and not judged: they are a second definition of open, which fails bags with an identical 83-88 mm mouth depending only on where above the carton their gussets draw in (the seated carton forces the body open where it stands).
- *Start state:* the candle starts on the counter in front of the mat on the centre line (`scene.CANDLE_XY_PURCHASE`, 0, -0.07 m; open-bag keeps it left of the mat). A Franka Hand opens to 80 mm and the carton is 90 mm across,

so it is held between two hands; left of the mat the right arm's IK reaches neither pair of its side faces at any of seeds 0-9, on the centre line both arms reach both pairs at every seed (static check, no collision or path test). Seeds move it ±20 mm, ±30° (at least 8.9 mm from the mat and 13.9 mm from the customer edge over seeds 0-9999).

- *Hands-off window:* the agent is not told to wait after letting go, so the explicit evaluation (GAR `evaluate()`: `/v1/evaluation`, `gar verify`, the benchmark) completes it: if the episode's last 3.0 s were not already free of robot contact, the world is advanced for the rest of that time with the arms holding their last command, judged, and restored exactly (`env.GiftBagsEnv.final_evaluation`). A contact in that branch fails the window; the start state is not advanced. The cost is the simulated remainder at the counter's wall/sim (minutes); the session declares a 1,800 s client timeout and tells owner and policy how the result is judged. The per-action owner reward is the same evaluation without the branch. A branch is a pure function of the episode's state only on one solver thread: the evaluation reports `solver_threads` and `bit_reproducible`.
- *Release window:* it counts robot contact only with the job's items: a touching robot link counts unless it is at least 20 mm from the tote (sheet 0, its ropes and knots), the candle, every board more than 20 mm from its start pose (`Evaluator._job_contact`) and any stock that is itself within 20 mm of a job item (pushing the job through a leaning lower tote counts). A hand resting on the remaining stock or the board pile does not fail it. `base_at_rest` reads the drift over the window's last 2 s at the same speed ($\leq$ 1.33 mm): a bag that shifts once as the carton lands at the start of a window completed from the release, then stands still, passes as it does when the agent waits (a recorded release from the robot's held tilt shifted 3.86 mm in 0.36 s); a bag still sliding fails. In progress, "on the board" is resting on any board weighted by that board's credit (`candle_on_board_credit`).
- *Score:* 1 on success, else 0.9 × progress. Progress (`_bag_progress`) reads the judged state only: a third each for the bag ((opened + opened × standing on the mat + squared)/3; opened = the smaller of the mouth ratio's ramp from the start state's value to 0.80 and the least rim gap's ramp from the start state's to G/2; standing mouth-up on the mat with its lowest point within 5 mm of it; squared = mouth up and resting within 5 mm on what it stands on, mat or counter, times its bottom's flat fraction against that support to 85 % times its tilt), the board (1 if it passes `board_in_bag`, else ½ its share inside the tote's hull + ½ its share counted by the board rule) and the candle (in the bag by its share of corners inside, × (1 + upright + upright × on the board)/3), everything but the bag's openness gated by the bag being mouth up (1 within 10° of vertical, 0 on its side or inverted), times the tote's integrity (1 up to 5 mm of new crack, 0 from 25 mm). Nothing is credited for order, path, grasp, arm, which board or for nothing having gone wrong (an untouched stack, a contact-free window); a nudged stack costs nothing, and an item still held fails success only (0.9 × progress). The components are in `metrics.reward`, the mean of the checks in `metrics.mean_of_checks`. Authored [A]: the equal thirds, the ramps' zero points, 5 mm (mat, board), 25 mm (tear), 0.9.
- *Controls:* constructed states (no physics) check the verifier: valid completions by any board, two boards, off-centre, stack nudged; the false positives named above; the progress ladder; damage; not released. Five recorded ASSISTED trajectories are judged by it as well. No robot or assisted run of the whole task passes: the only opening on record that squares the bottom (the ASSISTED mould of §8.2) tears the tote 79 mm on this world, and hand-sized holds do not square it.

Controls. The unmodified evaluator is run on recorded physics states: item poses from the recording, the paper midsurface rebuilt from the two recorded skins (the rebuild reproduces the live metrics of §6.7's factory-crease bag to three digits: mouth ratio 0.861, gussets 143.3° and 142.9°, tilt 0.109°, bottom 93.7 % flat). Each preset gets a positive control (the recorded state completed by one labelled edit, every check must pass) and negative edits applied to that positive state (one physical property pushed past its threshold).

Preset / task (recording)	State	Checks that fail
counter / open-bag (worked-crease tote standing on the mat, §6.7)	as recorded (no board inside)	`board_in_bag`
	positive: one board laid flat on the bag's bottom	none: **success**
	bag (and board) tilted 15°	`standing`, `bottom_squared` (the bottom tilts with the bag)
	bag (and board) moved 0.30 m off the mat	`on_mat`
	boards moved 0.5 m away	`board_in_bag`
	faces pushed to 25 % of the depth at the rim	`mouth_open`

Preset / task (recording)	State	Checks that fail
counter / open-bag (factory-crease tote after 10 s, §6.7)	as recorded	`board_in_bag`
	positive and the same four edits	none; then `standing` and `bottom_squared`; `on_mat`; `board_in_bag`; `mouth_open` and both `gusset_*_deployed`
home / lift-lid (slow lid lift, §8.1)	as recorded (lid held 60 mm up)	`lid_on_table`
	positive: lid set down flat 0.2 m in front of the box	none: **success**
	lid back on the box	`lid_off`, `lid_on_table`
	episode base lift 6 mm / tilt 11°	`base_not_lifted` / `base_not_tipped`
	frame taken out of the base	`frame_in_base`

8 Assisted demonstrations

Scope: labelled phantom fixtures (amber proxies in the videos) that show the physical interactions work. They are not policy actions and are unreachable through GAR.

8.1 Lid air film

A phantom hand (spring 5 kN/m, 40 N cap on the lid's centre of mass) lifts the lid 60 mm. At 10 cm/s the slot flow sucks the base (with the 350 g frame inside) up 1.3 mm off the table for ∼0.2 s (peak 193 Pa, 4.5 N > 4.05 N weight); at 0.5 cm/s the film force stays at 0.22 N and the base does not move (0.01 µm).

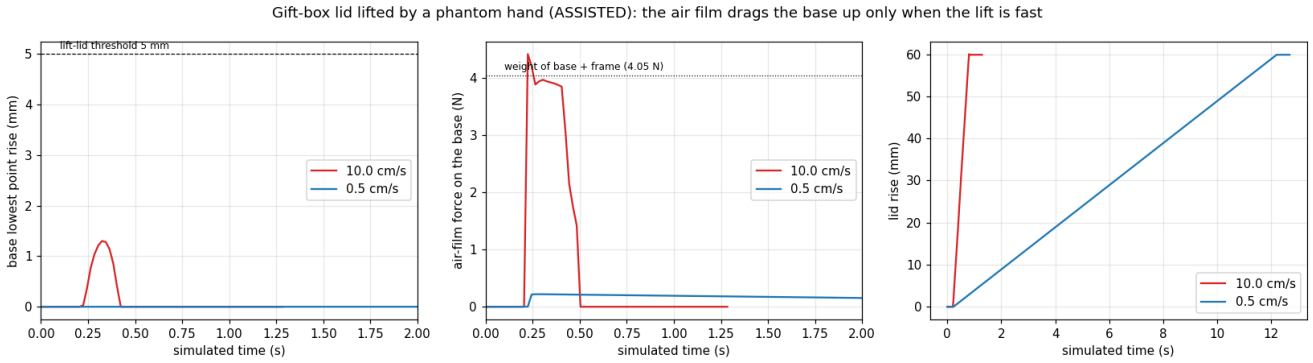


Figure 15. The ASSISTED lid lifts at 10 cm/s and 5 mm/s: lid and base rise and the air film's force.

8.2 Opening the flat tote

With the holds of two hands: not achieved. An ASSISTED bench opens the tote with what a clerk's two hands apply and nothing else: never more than two holds at a time, each a hand-sized patch of paper held by capped springs to where it was when the hand closed (20 N per hand in all), or a rigid fist-sized block, 90 x 70 x 30 mm with rounded edges, pushed by a capped spring (10 N while it slides in, 40 N when it presses). No hold knows the bag's open shape. The runs start from the shipping state of §6.4, which has no damaged triangle; each stage was run from the saved state of the one before, so the stages are not one continuous run.

Stage	What the hands do	What the bag does (measured at the end of the stage)	Largest hold force	Splits
lift	hand 1 pinches the middle 90 mm of the top bag's band, both faces together, lifts it off the stack, turns it upright over the mat	hangs 1.3° off plumb, flat (mouth ratio 0.12)	5.1 N	0

Stage	What the hands do	What the bag does (measured at the end of the stage)	Largest hold force	Splits
flap	hand 2 pinches the tip of the folded-up bottom flap and swings it down; lets go	the flap hangs down; bag 0.1° off plumb	4.4 N	0
open	hand 1 keeps the front band, hand 2 takes the back band; they move apart one gusset width	**the mouth opens fully** (ratio 0.99); the gussets open at the top only (median opening 25° and 16° of 180°); at their bottom edges the faces are 0.6 mm apart	3.2 N each	0
fist	hand 2 lets go and goes in on edge along the held front face, down to the faces' bottom edges, and turns flat	the faces part to about 77 mm at the fist and come together under it: 44 mm apart at their bottom edges on average, 37 mm at the middle; the bottom hangs below them as a closed V 45 mm deep; the fist is 3.9 mm short of the point it was sent to (6 mm above the faces' bottom edges); bag 6.0° off plumb	fist 10 N (its cap), hand 1 12 N	1, at a rim corner (0.14 mm counted)
down	hand 1 and the fist lower the bag onto the mat, the fist over the bottom's crease; the fist presses 1 s and comes out	**the bottom does not square**: its crease lies on the mat along the middle only, the half behind the crease stands at about 58°, both ends stay 32-47 mm up, folded in the gusset feet; faces 57 mm apart at the bottom (open: 102); bottom flat over 1.7 % of its area (limit 85 %); the bag leans 19°; mouth ratio back to 0.31	fist 40 N (its cap), hand 1 19.9 N (its cap)	the same one, now open over one rim cell (10.4 mm counted; `bag_not_torn` allows 5)
fist and down again, the block turned so its 90 mm lie across the bag's depth	as above	in the air the faces are 90 mm apart at the block and 37 mm at their bottom edges, the bag 13° off plumb; on the mat, saved while the block presses: **the block's underside is 23 mm above the mat, 30.7 mm short of its target; the bottom lies flat against it at that height with a pleat hanging from its middle down to the mat**; faces 58 mm apart at the bottom	fist 40 N (its cap)	1, at a rim corner (0.34 mm)
fist again, the hand sent on down into the V, two runs	as above	on edge the hand gets to 24 mm above the crease, inside the V. Turned flat about its centre (first run) it lifts itself out again. Turned about its lower edge, so that it should come down 30 mm as it turns (second run), **it is pushed up out of the V instead and ends 42 mm short**: the faces close under it as before, 43-45 mm apart at their bottom edges, the V shut; bag 16-21° off plumb	fist 10 N (its cap), hand 1 13 N	2, at rim corners (0.5-0.6 mm)

Where the bottom of a hanging tote is. It is not at the faces' bottom edges: it hangs 45-50 mm below them, a closed V. A hand turned flat at the faces' bottom edges turns above the V, the faces close again under it, and on the mat the block bridges the V's top and stands on it: pressing harder only pins the pleat. A hand goes down into the V and opens it from inside. The last row of the table sends the hand there: on edge it does get into the V, and when it turns flat with the 10 N it is allowed while sliding, the closing faces push it back up and out. **So these runs establish a force: 10 N on a rigid 90 mm block does not open the bottom's V from inside; the bag squeezes the block out.** Whether a real tote of this paper would let a hand in at a few newtons is not known; what holds the V shut here is its two ends in the gusset feet, and no run has worked on those. Not tried: the same turn with the press's 40 N; a hand that opens each gusset foot first.

What these runs establish, from the numbers and from sections drawn out of the saved states:

- *The gusset feet do not tear when the bag is opened by hand.* Through lift, flap, opening the mouth and the hand going in there is no split at a gusset foot and none anywhere with the faces pulled apart. The one split here is at a rim corner beside the held band and opens when the bag is pressed down crooked with hand 1 at its force cap.
- *Parting the bands opens the mouth and leaves the lower half closed.* That much behaves like a tote: the next move is the hand inside.
- *Why the folded bottom resists a fist*, as far as these runs answer it. In the air the fist parts the faces where it is and not below it: under it they come together again, about 40° off the vertical, onto a bottom that is still a closed V. What holds the V shut is its two ends, folded up inside the gusset feet; a block 90 mm long in the middle of a 203 mm bottom reaches neither. Set on the mat and pressed (the fist at its 40 N cap, so stopped short of its target; how far short was not recorded) the V opens to about 104° instead of 180°, lopsided, and the

ends stay up. The step that is missing is the one a hand makes next: into each corner.

- *Sequences that fail for the fixture's reasons:* the bag set down before the hand goes in folds its closed bottom over and is pressed flat folded double; a sharp-cornered block pushed in at 40 N tears the bag on the way down; entered as a block instead of on edge it comes down on the back wall's rim; pressed 9 mm off the crease it flattens one half only.

So the counter tasks are not shown to be completable with hand-like holds (§13).

Fixture openings from a bottom-extended start. The runs in this part start from the bag laid flat with its bottom panel unfolded, not as it ships, on a coarser mesh (1,731 vertices), and use a friction model in which a contact loaded below 2.5 times its limit could not slide (§3.2); their numbers are indicative. Phantom fixtures, labelled in the video and drawn as amber proxies, open the factory-flat top tote; the paper model decides everything below the gripped points (gussets unfolding, crease springback, the bottom squaring, standing or falling). The robots hold still. Six attempts:

Attempt	Fixtures	Outcome (hands-off window)
1	two mouth pinches (2 kN/m, 6 N cap per vertex) lift, open and lower the bag; a 6 mm plate presses from above	bottom stayed folded; bag tipped over on release
2	pinches + a hand-sized phantom block (170 x 70 x 40 mm, 40 N cap) aimed at the mat centre	the block went down outside the front face
3	pinches + the block aimed down the hanging bag's own axis	in the bag; bottom 73 % flat, gussets 9–13°; the bag fell onto its back 0.4 s after release (tilt 89°)
4	after the carry a phantom frame on all 60 rim vertices (1 kN/m, 1 N cap) pulls the mouth into its open rectangle; the bag is lowered in the frame, the block presses the bottom while the frame follows the last 10 mm; frame released at once	the bottom flap never squared and the bag stood 19 mm taller than an open bag, so the frame held the rim down with its full 60 N; on release the buckled walls sprang back, the bag leapt ∼13 cm, somersaulted and landed flat (tilt 85°, gussets 56° / 6°)
5	rim frame as in 4, held above the mat; a bottom former (phantom block 190 x 88 x 60 mm, the bag's inside less 6–7 mm a side) goes in through the open mouth and pushes the bottom to the open bag's depth; frame and former set the bag on the mat without pressing; the former withdraws; the frame's springs fade out over 0.5 s	the former never entered the bag: its body stayed on the mouth, stopped by the rim, while its target went down inside; the bag hung 307 mm tall with its bottom flap extended, crumpled against the mat when lowered and fell 0.3 s after the frame let go (tilt 88°, gussets 8° / 98°)
6	rim frame as in 4; a second spring set on the bottom panel's 231 vertices (1 kN/m, 0.5 N cap each, 115 N in all) pulls the bottom toward its square, flat open position under the held rim; both set the bag on the mat, hold 1 s and fade out over 0.5 s	the bottom panel never got flatter than 53 mm top to bottom and the bag stood 283 mm tall instead of 254 mm: holding the rim and the bottom does not drive the folds through the snap that turns the flattened bottom into a block bottom; the bag fell 0.6 s after release (tilt 86°)

A mould. An ASSISTED mould asks the world directly: a capped spring on every one of the tote's 1,731 vertices carries the flat bag from the stack into its open shape on the mat over 2 s, holds 1 s and fades out; then nothing touches it for 3.2 s.

Mould	Self-contact	Mean / max distance to the open shape at the end of the hold	Bag height (open: 254 mm)	Bottom panel	After release
200 N/m, 0.05 N cap per vertex (87 N in all)	on	40.8 / 76.1 mm	285 mm	folded	fell, tilt 89°
200 N/m, 0.05 N cap	off (diagnostic)	36.6 / 74.7 mm	243 mm	folded	fell, tilt 84°
2 kN/m, 0.5 N cap (866 N in all)	on	17.1 / 67.5 mm	256 mm	45 mm out of flat; 80 % of its crease edges still bent > 30°	fell, tilt 89°
2 kN/m, 0.5 N cap	off (diagnostic)	22.4 / 67.5 mm	247 mm	52 mm out of flat; 68 % of its crease edges bent > 30°	fell, tilt 88°

Every fixture and both moulds bring the walls and the mouth close to open but leave the block bottom folded: its bottom-centre and diagonal crease lines do not unfold, and 13–19 % of the bottom panel's plain edges kink by more than 30° instead. Switching the tote's self-contact off does not help, nor does an eight-times softer glued bottom (bottom 43 mm out of flat) or ten-times softer creases (42 mm). §6.7 shows the open state itself stands.

What stops the mould is its own path. Each vertex's target moves on a straight line from the flat bag to the

open shape. Measured on the mesh, those intermediate targets are not paper shapes: halfway along the ramp 74 % of the tote's edges would have to shrink by more than 10 %, many to zero length, because half of the flat bag's layers face the opposite way from where they end up. Paper cannot shrink, so it crumples; the recorded frames show the lower half crumpling from 0.8 s on, and sliced at the end of the hold the bottom corners are rounded over ∼50 mm and the lower gussets drawn in by up to 50 mm (the mould section figure under Further figures). The crumples are not what holds it: with the panels' plastic hinges switched off (a diagnostic) the bottom still ends 40 mm out of flat and the bag falls, so even perfectly elastic paper is pulled into the wrong folded configuration. The straight-line path demands the same compression at any mesh resolution, so a finer mesh cannot answer the question. With the in-plane stiffness lowered a hundredfold (a diagnostic, not paper) the same mould brings the bag into the open shape to 0.11 mm mean and a 0.8 mm flat bottom (it then collapses after release, being floppy). So the obstruction is the paper's inextensibility along that path. The evaluator's positive control (§7) shows the goal is well defined and reachable as a state.

Opening along the bag's own folding motion (ASSISTED). The fixture moves the bag only through shapes paper can take: a grip on every vertex carries the flat bag off the stack and turns it upright over the mat (a rigid motion, largest edge strain 0.002 %); then springs on the held vertices pull the front and back faces apart along the exact folding motion of a gusseted tube (faces translate, each gusset half turns about its corner crease; 0.0 % edge strain); the bottom panel and the gussets below their diagonal folds are free; the held walls set the bag on the mat, hold, fade out over 0.5 s, and nothing touches it for 3.2 s.

Held	**Bottom out of flat (end of hold)**	**Mouth ratio**	**Gusset opening (median; length ≥ 120°)**	**After release**
faces and gussets above their diagonals (1,452 vertices)	19.6 mm	0.98	163°; 72 %	stood ∼1 s, the bottom sprang back 12 mm, toppled (88°)
walls as above, then a hand-sized former (196 x 70 x 30 mm, 40 N cap) presses the bottom, the board is laid in and pressed	25.6 mm	0.97	160°; 72 %	the former stopped 96 mm above the mat and the board hung at 87 mm, both resting on the drawn-in lower gussets; the bag toppled (87°)
as the previous row, with the whole gussets also held flat once on the mat	19.8 mm	0.97	161°; 72 %	held gusset vertices still up to 21.7 mm off target; board again not on the bottom; toppled (87°)
DIAGNOSTIC: as the previous row, the bottom-perimeter creases given the open shape's 90° as plastic memory when the opening starts	21.6 mm	0.94	149°; 96 %	**stands** (2.7°, drift 0.65 mm); fails only the squared bottom (5 % within 3 mm) and the board (none in this run)
same DIAGNOSTIC, with former and board	22.4 mm	0.96	159°/150°; 96 %	**stands** (3.5°); fails the squared bottom (5 %) and the board (6.7 % on the bottom)
only the faces' 40 mm turn-top bands (168 vertices, where two hands hold)	50.2 mm	0.30	37°; 0 %	did not open; fell (88°)

Along a path paper can follow the walls-held opening gets far closer than any other fixture (the bottom lies on the mat over the whole footprint with ∼10 mm ripples; the gussets' lowest ∼50 mm stay drawn in by up to ∼15 mm), but once released the bottom's factory creases spring partly back and the bag topples. The rim bands alone do not open it. In the real procedure the clerk presses the bottom square from inside and drops the bottom board in. The former and board rows add exactly those steps, but the former and the board come to rest on the gussets' lowest ∼110 mm, which stay drawn inward because the 0.5 N springs cannot hold them flat (held vertices up to 21 mm off target). Holding the whole gussets flat once on the mat, as a clerk runs fingers down them, does not change that. A state dump on the mat shows what holds the gusset feet in: the faces' bottom edges sit 15–19 mm above the mat and are pulled up to 9 mm inward while the bottom panel bows down onto the mat at its centre, i.e. the bottom-perimeter creases spring back toward flat. In the bottom-extended start those scores lie flat, so their plastic memory is 0°; a manufactured tote's perimeter scores are formed folded and the tote ships with the flap folded up. Giving only those 60 crease hinges the open shape's 90° as memory (a diagnostic) is enough for the opened bag to stand after release with mouth and both gussets passing (rows above); the bottom is still not flat enough and the board still does not reach it.

From the shipping state. The task's tote starts as it ships, flap folded up, pressed and stored, on a mesh graded to 0.8 mm at the bottom folds (§6.4). The staged opening has one more step for the folded flap, placed where a person does it: the bag hangs upright, held flat, and a phantom pinch on the middle of the flap's tip swings the flap

down about the fold line before the faces are pulled apart. Then as before: walls pulled apart in the air, the former pushes the bottom out, the walls set the bag on the mat, the former holds the bottom flat 3 s and withdraws, the gussets are smoothed, the board is carried in tilted through the turn-top band, levelled and pressed, everything fades out, 3 s hands off. **What holds the bag in these runs is a mould, not two hands:** capped springs (2 kN/m, 0.5 N each, 1,380 N in all) on 2,760 of the tote's 3,559 vertices, each driven along the ideal folding motion of a gusseted tube. The rows below say what that mould reaches; what two hands reach is the first part of this section. These runs use an earlier build of the shipping state, whose fold had left crushed triangles at the gusset feet (below), and the earlier friction model; on the current world the same mould opening tears the tote 79 mm (§7).

Run	Stands	Mouth	Bottom flat (limit 85 %)	Gussets (limit 120°)	Board on the bottom	Torn (limit 5 mm)	Evaluator
DIAGNOSTIC, no rope handles	yes, tilt 1.9°, drift 0.13 mm	0.92	**83.8 %**	155° / 151°	**0 %**: it ends outside the bag	**122 mm**	fails 3 of 11 checks
With rope handles threaded through their holes (film)	yes, tilt 0.66°, drift 0.06 mm	0.91	**100 %**	148° / 151°	**100 %**	**178 mm**	**fails 1 of 11 checks:** `bag_not_torn`
The same with the candle put in after the board (`bag-purchase`)	yes, tilt 0.23°, drift 0.002 mm	0.91	**100 %**	145° / 148°	**100 %**; candle inside (8 corners), upright (0.4°)	**155 mm**	**fails 1 of 13 checks:** `bag_not_torn`

With the handles attached the opening reaches the task's goal state in every respect but one. In the second row the tote stands on the mat with its mouth and gussets open, its bottom flat over its whole area, the board lying on it, the knots at their holes 25 mm under the rim and both loops hanging outside the faces; ten of the evaluator's eleven checks pass. The eleventh fails: the bag has torn 178 mm (below). 27 s simulated at 204x wall/sim, 190 of 6,950 steps at the iteration cap.

What the run without handles shows, and what it does not:

- *The tote opens from its shipping state into a standing bag with open gussets.* From the bottom-extended start it toppled or needed its crease memory rewritten. The folded-up history is what a manufactured tote has, and with it the bottom-perimeter scores hold their fold.
- *The bottom is 1.2 points short of flat enough* with the former's 80 N held 3 s.
- *The board did not go in, and that was the fixture's path, not the bag.* The phantom grip carried the board on a straight line from 10 cm above its stack to a point over the mouth. Read from the recorded board poses, its leading end met the bag's left wall 59 mm below the rim at t = 15.0 s and stayed there (x = -181 mm) while the capped grip's target went on into the bag; when the grip let go the board fell to the mat outside, upside down, 220 mm left of the bag. In the film the yellow plate is the grip's target, which does enter the mouth; the board itself is the grey plate left behind. In the runs with handles the path lifts the board above the rim first, then carries it across.
- *The bag tore 122 mm* in 19 splits within half a second (t = 3.8-4.3 s) at the two ends of the fold line, along the finest mesh rows at the foot of both gussets, while the phantom hands pulled the faces apart (not while the flap was swung down, which ended at 2.9 s). The runs with handles tear in the same place by about the same length.

Where it tears, and why. In a run with handles fifteen of its eighteen splits run up the foot of each gusset, at its centre crease and one cell either side, from the bottom edge to 7 mm above it, all within 0.16 s near t = 4.0 s; the other three are at the rim's corners. That is where the shipping fold of that build had crushed the paper: the fold line's two ends, where four joined layers turn the fold on different radii. Of the 216 triangles within 6 mm of the split vertices, one in ten carried an in-plane set above 24 % from the fold, the largest 55 %; in the whole bag 172 of 6,336 triangles were set by more than 5 %. As the bag is opened those triangles are strained against the rest shape that set gave them: up to +34 % and −36 % by the end of the flap's swing, +22 % and −65 % at 3.92 s, just before the splits, while 99 % of the bag's triangles stay within 1 %. That 55 % set was not a property of paper: it was made while the tote's rest shape was still a blend of the authored and the relaxed fold, with the crack band running. The current generator damages and splits nothing while a rest shape is a blend, and paper crushed in its plane keeps its true length as slack, which pulls out under little force (§6.4).

The rim-corner splits are started by the tear-start rule's out-of-plane measure. Across a closed fold a normalised

mean of the two leaves' normals is arbitrary, and parting the leaves of a fold would read as a trouser pull, so the mean is not normalised (it vanishes across a closed fold). The evaluator's crack length counts a split edge by the fraction of its bond's fracture energy that was dissipated, so a split that never opens counts nothing (§6.6, §7). The two runs with handles have three such splits each, one of them at a back corner, and none of their bonds ever opened.

The counter tasks are not shown to be completable (§13): both openings reach the goal state, `bag-purchase` with the boxed candle standing upright on the board (a phantom hand lifts it, carries it over the mouth and lowers it in; 32 s simulated at 196x), and both tear the bag on the way; no robot has done any of it.

8.3 Unwrapping the present

At each end of the wrapped present the bottom flap (a trapezoid narrowing from the full end width to $\sim \pm 25$ mm at its top edge, 42 mm above the box bottom) is folded up over the end face and held by a 19 x 50 mm strip of Magic tape straddling its top edge. Phantom fixtures, labelled in the video, open it: a fingernail lifts the tape, a thumb and a finger swing the end's side flaps out, and a second phantom hand (a 5 kN/m, 40 N spring on the base tray's centre of mass) holds the box down, as a person steadies a present. Whether the tape peels (cohesive law, Γ 150 J/m^2), the paper gives way or the flap only bends is left to the models.

The whole unwrap, stage by stage (ASSISTED). **It is not shown: the present has not been unwrapped to the task's criterion by tearing or by sliding.** A script runs the stages in turn, measures a criterion after each and stops at the first that fails. Nine attempts. The tape and end stages were run once per opening angle and their states saved; the later rows continue from those saved states, so each record below is a continuation and is not one continuous run.

Tape. The fingernail lifts one end of the end tape and draws it off the top flap: none of it is left bonded there, 94 % of its area on the bottom flap still is (nail 7.2 N at most, box hand 5.6 N). Passed.

End. The thumb and finger swing the two side flaps out as doors. How far they are swung decides what the end looks like afterwards (angles as the script measures them: side flaps from the end face, 90° is straight out; top and bottom flaps from straight out, 90° is folded on the end face, negative is past straight):

Side flaps swung to	Side flaps reach (back / front)	Top flap	Bottom flap	Then	Result
120°	106° / 109°	11°	32° up	the box hand slides the box out, thumb and finger holding the side flaps	**FAILED**: the box moved 29 mm and the paper 27 mm with it; then the box hand sat at its 20 N cap
120°	as above			a palm (15 N on 33 vertices) presses the bottom flap flat first	**FAILED**: the pressed part came down from 20.7 mm to 7.4 mm above the table; the criterion is 3 mm
120°	as above			the paper is held on the box's top near its far end (15 N) instead of by the side flaps	**FAILED**: the box moved 35 mm and the paper 30 mm with it; the hold was at its cap
120°	as above			tearing: a hand takes the top flap and rips the paper back over the box	**FAILED**: 24 splits, 189 mm of new tear, the paper's coverage of the box fell from 82.9 % to 73.2 %; the stage asks for 15 points and the task for 5 % in all
90°	85° / 85°	−25°	4° up	palm on the bottom flap	**FAILED**: 10.1 mm → 6.6 mm (criterion 3 mm)
90°	as above			slide, thumb and finger on the side flaps	**FAILED**: the box moved 3.7 mm and the paper 2.1 mm; box hand at its cap
100°	96° / 94°	−10°	7° up	slide, thumb and finger on the side flaps	**FAILED**: the box moved 20.0 mm and the paper 17.2 mm with it; box hand at its cap; the bottom flap ended 73° up
100°	as above			the fingers let go; the paper is held on the box's top near its far end (15 N)	**FAILED**: the box moved 33 mm and the paper 30 mm with it; the hold was at its cap, the box hand at 18 N. Let go, the side flaps swung back to 59° and 66° and stood in the box's way

Side flaps swung to	Side flaps reach (back / front)	Top flap	Bottom flap	Then	Result
100°	as above			the same with a firm two-handed pull: up to 40 N on the paper and 40 N on the box	**FAILED**: now the paper stays (0.6 mm) and the box jams after 5.9 mm with its hand at the 40 N cap, the hold at 34 N; the box rides up 8.5 mm; side flaps at 64° and 75°

What these runs show, read from the numbers and from drawings of the saved states:

- *The paper does not clamp the box.* The box moves 20-35 mm under less than 20 N.
- *The paper goes with the box.* In the three slides that went anywhere (20-35 mm) the paper over the box travelled 86-92 % as far as the box did. Holding the side flaps by their free edges does not hold the sleeve back: the flaps point along the slide, so when the sleeve follows the box they only buckle (with the flaps at 100° the thumb and finger read 2-9 N each while the paper travelled its first 15 mm, against caps of 16 N). The hold on the top paper near the far end did pull, and was overrun at its 15 N cap. That is the fixture's fault. A person pins the paper to the table or grips the sleeve itself.
- *Held firmly, the paper stays and the box jams in the opened end.* With 40 N on the paper and 40 N on the box the paper moves 0.6 mm and the box 5.9 mm, then nothing, with 34 N on the hold. Drawn from the recording: once the fingers let go, the back side flap swings back to about 64° and, with the tuck that ties its lower edge to the bottom flap, makes a web across the lower back corner of the opening, some 40 mm along the table and 35 mm up the side; the box's corner runs into it and the box's front rides up 8.5 mm; the top flap has tented 38 mm above the lid. The front side flap and the rest of the opening are clear. A person holds that flap open, or flattens it, while the box comes out; this run let go of it in order to free a hand for the paper. Whether the web would give way on a real present at a few newtons (the paper crumpling at the corner) and does not here because the wrap's 8 mm cells cannot crumple that finely was not tested and is not claimed.
- *Tearing works as a mechanism and was not carried far enough.* The paper tears at hand forces (24 splits, 189 mm, with 12 N on the top flap) but that pull bares a tenth of the box. An unwrap by tearing needs the paper torn along the box's length and lifted away, or the box lifted out of it, and no such sequence was scripted. The instruction of `open-present` allows picking the present up (§7), and `unwrap-gift` does not forbid it.

So unwrapping the present with assistance is **not shown**: the fixture sequences have not held the paper where a person would. The route that remains untried is the one people use: pin the paper to the table beside the opened end with one hand and draw the box out with the other, or tear the top open end to end and lift the box out.

9 Robot demonstrations (unassisted, scripted)

Scope: scripted joint-target and gripper commands through the same `GiftBagsEnv.act` path a policy uses (IK computes targets only). These show the robot coupling working on the real items; they are not policies.

9.1 Picking up an opened tote by its rim (UNASSISTED): achieved (a component check, not a task)

Start state (labelled, not a policy action): the top tote stands opened on the mat with worked creases and without its rope handles, as in §6.7 (the knots and loops are not placed for the open shape). Everything after that is robot commands through `GiftBagsEnv`: the right FR3 raises its open hand beside the bag, crosses above it and comes down over the back wall on a Cartesian path checked before anything moves, lowers it so one finger is inside the bag and one outside, closes in force mode (20 N per finger) on the 40 mm turn-top band, lifts 10 cm, holds 1 s, sets the bag down and lets go.

Result: the bag's lowest point is 78 mm up at the end of the lift and 94 mm up at the end of the hold (100 mm commanded); the contact world reports 15.3-16.6 N on each finger link through the lift, hold and set-down (no chatter); the bag stands again at 0.1° with its bottom flat; no velocity reflex, no paper damage; 5 of 2,900 steps at the iteration cap; 59x wall/sim. The evaluator is not the measure here: the run ends 0.5 s after letting go and

there is no board.

The approach is a checked Cartesian path: a first move interpolated in joint space from the ready pose sweeps the hand through the standing bag, and a straight Cartesian rise from the ready pose is refused as an IK branch change, so the hand first rises where it is. Grasps between the finger pads are resolved inside the implicit contact solve (§4), which is why this one is steady where §9.3's squeeze is not.

9.2 Taking the flat tote off the stack (UNASSISTED): not achieved

The right FR3 presses its closed fingertips on the top bag near its bottom end and drags it toward itself (pad on laminate μ 0.8 beats laminate on laminate 0.35) so the mouth overhangs the stack, then turns its fingers horizontal to pinch the overhang. Two runs. In the first the 15 cm lift was out of IK reach once the force-mode grasp was made. The second, with every waypoint checked beforehand: the drag slid the top bag 12 mm (60 mm commanded; the press force chattered between 5 and 32 N); while turning the fingers horizontal the IK chose another arm branch, joint 1 swung ∼1.9 rad, the arm swept away from the stack and tripped the velocity reflex 18 times, and the fingers never reached the bag edge; the lower finger pushed the stack 13 mm. A script failure (branch selection), aggravated by the contact force spikes of §13 item 7.

The demonstrations solve every motion as a continuous Cartesian path before simulating (`ArmKinematics.cartesian_path`: poses at most 1 cm and 8° apart, each solved from the previous solution, any joint jump over 0.35 rad refused). For this pick the check refuses the path where the run failed (turning the fingers horizontal beside the mouth jumps branch). A search over twelve hand yaws for the drag (0–330°) and grasp approaches tilted 0–45° below horizontal, both closing-axis signs, found no continuous path: every candidate failed with a joint jump or an unreachable pose while turning the hand toward the grasp. The overhanging mouth lies 12–20 cm in front of the right arm's base, at the base's own height. A clerk would take the bag by its rope handle, which is modelled and holds the tote's weight (§6.4) but has not been tried by a robot.

9.3 Lifting the lid with two hands (UNASSISTED): achieved at 1 cm/s; fails at 10 cm/s

Both hands close their fingers, descend beside the lid's short ends and squeeze it between the outer faces of their closed fingers (the lid, 184 mm long, is wider than the Franka Hand's 80 mm opening; 12 mm of position interference into each end), lift, carry the lid toward the front of the table, set it down, open outward and retreat; the base and the photo frame inside it are left to the air film and contact.

Robot model on the CPU, Coulomb friction (§3.2).

Lift	Solver threads	Evaluator	Base: lowest point / one end, peak rise	Base tilt	Base slide	Peak film force	wall/sim
1 cm/s (film)	2	**success** (score 1.0)	0.03 / 0.04 mm	0.000°	0.65 mm	0.17 N	22x
1 cm/s	1	**success** (1.0)	0.03 / 0.04 mm	0.003°	0.68 mm	0.17 N	69x
10 cm/s (film)	2	**FAILED** (0.67): `lid_off`, `lid_on_table`	0.29 / 1.87 mm	0.55°	2.98 mm	4.48 N	44x
10 cm/s	2	**FAILED** (0.67), the same two checks	0.03 / 1.98 mm	0.58°	2.72 mm	4.19 N	42x
10 cm/s	2	**FAILED** (0.67), the same two checks	0.22 / 3.96 mm	1.24°	2.61 mm	4.26 N	40x
10 cm/s (run a)	1	**FAILED** (0.67), the same two checks	0.03 / 1.52 mm	0.45°	2.76 mm	4.28 N	49x
10 cm/s (run b)	1	**FAILED** (0.67), the same two checks; every logged number equal to run a	0.03 / 1.52 mm	0.45°	2.76 mm	4.28 N	52x

At 1 cm/s the robot takes the lid off and the base stays down. At 10 cm/s the same scripted motion loses the lid in every run: the air film pulls the lid back with 4.2-4.5 N, the lid slips out of the squeeze during the lift and drops back onto the base, where it ends askew (28.5 mm of overlap, its lowest point 26.6 mm above the table). The base is not lifted past the limit in any of them (one end rises 1.5-4.0 mm against 5). This is the demonstration's failure, not the world's: a faster lift loads the grip through the air film, which is the point of the task, and a demonstration with margin would squeeze harder or lift slower.

Reproducibility. The two one-thread runs at 10 cm/s agree in every logged number, and so would any two runs on one thread: the contact world is bit-reproducible there. On two threads the three runs issue identical commands and differ in the base's rise (1.9 to 4.0 mm at one end) and tilt (0.55° to 1.24°): the parallel linear solves are not

bit-reproducible, and a lid slipping out of a grip amplifies the last digit. The verdict is the same in all of them here; near a limit it need not be.

The right hand's orientation. The arm's inverse kinematics can return a configuration in which the wrist runs into the arm's own forearm (the IK does not check self-collision, §13 item 18). The right hand then stands 17-22° off the orientation it is commanded (1.1 kN of self-contact), and the skewed finger face pushes the box 10.2-10.7 mm sideways and turns it 4.7°. The right hand is therefore turned half a turn about its tool axis, which reaches the same grasp without the collision (orientation error 0.13° before the squeeze, 1.5° while squeezing, no self-contact), and the demonstration records both. The base slides 0.65-0.68 mm at 1 cm/s and 2.6-3.0 mm at 10 cm/s.

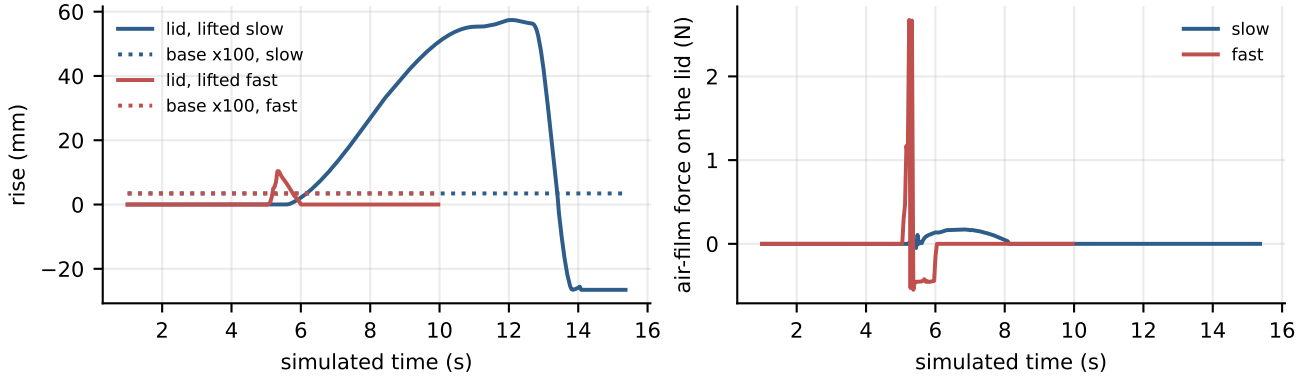


Figure 16. The 1 cm/s run and the first 10 cm/s run (the one that lost the lid), from their records.

The squeeze. The instrumented squeeze bench (every 4 ms step logged) holds 19.6 N mean (σ 1.0 N, no step under 1 N) through a 2 cm lift after an initial 0–40 N transient that decays within ∼0.6 s. Each move interpolates from the previous command, not from the measured joints: those lag the commanded ones under load, so a move that started from them would begin by releasing part of the squeeze.

Reactions are read from the coupling side: the force the obstacle's spring, finger drive and finger inertia transmit, which is bounded by the caps the robot feels. At equilibrium it equals the reaction read from the barrier gradient (median difference 0.0 N over 852 contact steps of the squeeze bench, lift grip 19.56/19.57 N, lid rise 18.88 mm). The finger's cross-axis coupling spring is capped like every other link. In the runs of the table the largest finger reaction is 42 N.

9.4 Opening one taped end of the present (robot motion, ASSISTED grip and box hold): not achieved

A person lifts the taped edge of the end flap with a fingernail and swings the flap down and out about its bottom crease. A Franka Hand has no fingernail and cannot pinch a flat flap on a vertical face, so the grip is ASSISTED: the flap's outer-layer vertices under the tape are held to points fixed in the right hand's TCP frame by capped springs whose equal and opposite load acts on the hand in the robot simulator. The grip is contact-gated (refused unless a finger surface is within 6 mm of the paper; the touch point is placed from the finger meshes). The arm's motion is the robot's: closed fingertips to the taped edge from above at 45°, a 70° swing along the arc about the flap's bottom crease (a continuous Cartesian path solved before simulating), hold, let go, retreat. A phantom hand holds the box, as a person steadies a present (ASSISTED).

With 5 N per gripped vertex and the box hold capped at 100 N, **the end did not open.** The grip was accepted 2.4 mm from the paper; the arm pulled with up to 53 N on its hand and the box hold reached 67 N; the taped edge came out 20 mm at most and stood 15 mm out at the end; 4.5 % of the end tape's bonds let go (54 mJ dissipated in that tape); no triangle tore; the box tipped up to 7.8° under the pull and stood upright again. The swing pulls the whole closure out as one piece and peels almost nothing. A fingernail peeling the tape back along itself takes it off at 7 N (§8.3); the Franka Hand has no way to do that. The film (the end in profile and from in front) ends with the closure standing out at the end, a slot open along its bottom edge through which the white box shows, and the tape still across the flap.

10 Stability: rest and legal abuse

Scope: does the world stay still when nothing acts on it, and stay finite and intersection-free under legal actions?

Rest. Each task's start state is left alone through `GiftBagsEnv` (robots holding their ready pose, no action); every 0.25 s the probe records the largest displacement of paper and of rope from t = 0, the paper's mean displacement (a uniform slide), the largest item displacement, kinetic energy and the Newton statistics, and it reports the rate of each over every 5 s window. Two solver threads; each run is filmed from its recorded states.

Preset / task	**Simulated**	**Paper: largest displacement; its rate in the first and in the last 5 s**	**Rope**	**Items: largest displacement; rate in the last 5 s**	**Newton per step / steps at the cap**	**wall/sim**
home / lift-lid	10 s	no paper	–	$1{\cdot}10^{-10}$ µm; 0	1.01 / 0	30x
home / unwrap-gift (and open-present: the same start state)	30 s	179 µm; 14.6 → 1.5 µm/s (uniform slide 0.46 → −0.02 µm/s)	–	8.9 µm; 0.03 µm/s	1.02 / 3	193x
counter / open-bag (and bag-purchase)	60 s	838 µm; 75 → 0.002 µm/s (uniform slide 4.7 → −0.001 µm/s)	245 µm; 22 → 0.005 µm/s	21 µm, 0.085°; 0.0006 µm/s	1.00 / 0	127x
receiving / open-carton	10 s	tape 0.002 µm; 0.0003 → 0 µm/s	–	0.002 µm; 0	1.00 / 0	54x
home / unwrap-gift, seeds 5 / 7 / 8 (another wrap stock; another box; that box with the tightest lid; 10 s)	10 s	growth of the largest displacement in the second 5 s: 36 / 12 / 91 µm; no split	–	growth in the second 5 s: 1.0 / 0.4 / 6.6 µm	1.05, 1.03, 1.02 / 3, 2, 2	123 / 184 / 189x
home / open-present (the same start state as unwrap-gift, another evaluator)	10 s	118 µm; 14.6 → 7.6 µm/s (the first two windows of the 30 s run, to the digit)	–	7.2 µm; 0.32 µm/s	1.06 / 3	107x

The tote on its stack comes to rest. It does so because of the stick displacement of §3.2: without it, a contact below its limit creeps at a speed proportional to its load, and the tote slides 1.9 mm in 10 s. With it the paper moves 0.84 mm at most in 60 s, nearly all of it in the first 10 s (the rope loops of the stacked bags are still settling), 1.6 µm/s in the next five seconds, and the rate then halves every five seconds down to 0.002 µm/s in the last five. No step reaches the iteration cap. The transition velocity is 1 mm/s.

The wrapped present comes to rest more slowly. Its paper moves 0.18 mm at most in 30 s; that largest displacement grows by 14.6 µm/s in the first five seconds and still by 1.5 µm/s in the last five (probably a wrinkle of the loosened wrap settling, 1 mm of slack, §6.5; it was not located). The box under it moves 8.9 µm in all and 0.03 µm/s at the end. Between the first and the last frame of the film 0.03 % of the pixels differ.

The lid box and the carton do not move (10^{-10} and 10^{-3} µm).

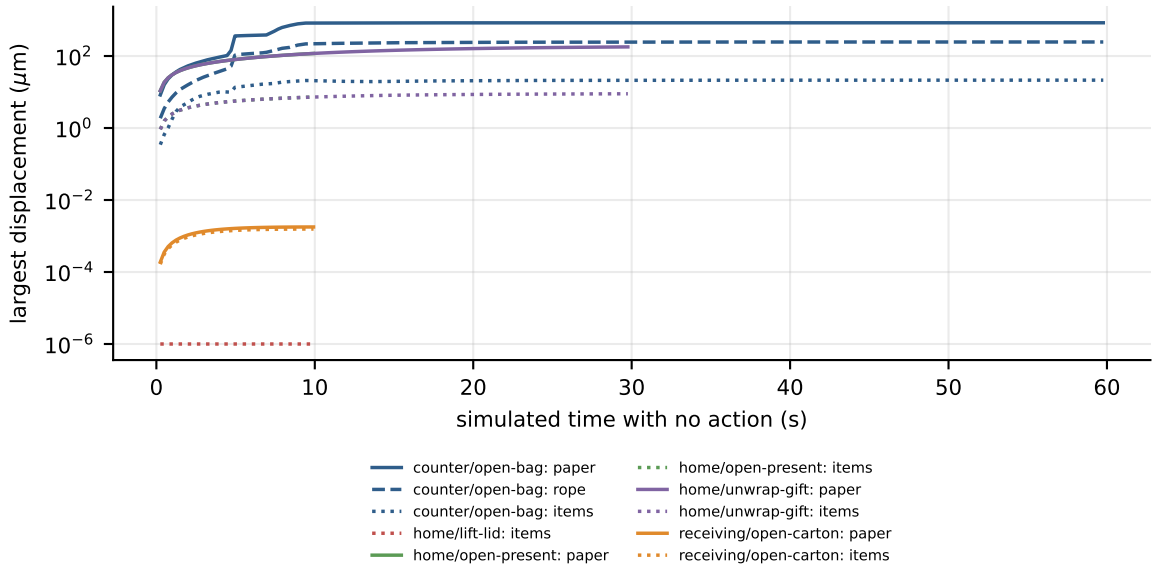


Figure 17. Largest displacement from the start state with no action (log scale); paper solid, rope dashed, items dotted.

Legal abuse. For 4 s, every 0.5 s, each arm is sent either a random tool pose near the task's items (within ±0.15 m of their centre, from 0.02 m below to 0.12 m above it, approach tilted up to 60°; IK gives joint targets only, so the hand can be driven into the paper, the items or the furniture) or, with probability 0.3, random joint targets within the limits; the grippers get random position or force commands. Each arm command runs as the public action path runs it: a minimum-jerk motion over its 0.5 s, **refused if it would ask a joint for more than its velocity limit**. Then the arms go back to their ready pose (1.5 s) and nothing acts for 3 s. Read after every step: finite state, intersections in the contact world, the fastest paper vertex, item and hand; at the end, whether what was pushed about has come to rest (two solver threads).

Preset / task, seed	Arm commands of 16: refused as too fast	Closest hand to the items' centre	Fastest paper vertex / item / hand	Largest force on a robot link from the contact world	Finite / intersections	Steps at the iteration cap	Velocity reflexes	At rest 3 s after	wall/sim
home / lift-lid, 0 (film)	12	0.09 m	– / 0.87 m/s, 5.6 rad/s / 1.35 m/s	156 N	yes / none	14 of 2,200	2	yes	30x
home / lift-lid, 1	11	0.08 m	– / 0.71 m/s, 5.9 rad/s / 1.76 m/s	251 N	yes / none	39 of 2,200	1	yes	32x
home / lift-lid, 2	11	0.12 m	– / 0.44 m/s, 2.5 rad/s / 2.04 m/s	16 N	yes / none	4 of 2,200	2	yes	19x
home / unwrap-gift (the wrapped present), 0	12	0.09 m	1.30 m/s / 0.65 m/s, 5.7 rad/s / 1.37 m/s; the paper moved 103 mm	251 N	yes / none	119 of 2,200	2	yes (paper 0.03 mm/s)	289x
home / unwrap-gift (the wrapped present), 1	11	0.06 m	0.91 m/s / 0.67 m/s, 3.3 rad/s / 1.79 m/s; the paper moved 92 mm	253 N	yes / none	223 of 2,200	0	yes (paper 0.03 mm/s)	242x
receiving / open-carton, 0	12	0.08 m	0.41 m/s (the tape) / 0.40 m/s, 0.7 rad/s / 1.62 m/s	253 N	yes / none	23 of 2,200	0	yes	33x
counter / open-bag (the tote with its handles on the stack), 0	12	0.22 m	the paper moved 0.09 mm; hand 1.52 m/s	0 N	yes / none	0 of 2,200	0	yes	57x

Preset / task, seed	Arm commands of 16: refused as too fast	Closest hand to the items' centre	Fastest paper vertex / item / hand	Largest force on a robot link from the contact world	Finite / intersections	Steps at the iteration cap	Velocity reflexes	At rest 3 s after	wall/sim
counter / open-bag, 0, **refused commands sent again with the least duration their refusal names** (10 arm commands, 6 of them re-issued)	6, all sent again	0.08 m	1.25 m/s / 0.82 m/s, 8.8 rad/s / 1.76 m/s; the tote's paper pushed 111 mm	184 N	yes / none	193 of 2,231	**179**	yes (paper 0.01 mm/s); no split	175x

On the counter the random commands alone do not reach the tote, and the row that does reach it shows a second thing about the rule. With 12 of 16 arm commands refused, the hands never came within 0.2 m of the flat tote: the first counter row is a null test and is reported as one. A client that is refused does what the refusal tells it, so the last row sends each refused target again with the least duration named in the message: the hands then reach the bag, push its paper 111 mm and the handles about, and the world stays finite, free of intersections and unsplit, and comes to rest. But the arms trip their velocity reflex 179 times: a motion that takes exactly its least duration runs at the joint's velocity limit, and the tracking overshoots it. **The refusal's least duration has no margin; a client that obeys it to the letter gets safety stops.** That is a defect of the action interface, not corrected (§13). The present and the carton were not run with re-issued commands.

What the rule costs the test, and why it is there. Two thirds of the random arm commands are refused: a random pose 0.3 m away cannot be reached in half a second within the joints' velocity limits. Without the refusal, with the IK poses sent as smoothed paths and the random joint targets as single steps, the hands reach 2.5-3.1 m/s and trip the velocity reflex 6-9 times a run. On seed 2 of the lid box the step targets throw the right arm: 65 reflexes in 0.5 s, one joint 0.15 rad past its limit, and the run dies on it. That is not a fault of the paper world but of an action interface that accepts a joint target as an instantaneous set point; the interface refuses one that is too fast, with the least duration in its message (§12), and the probe does the same. Runs without the refusal on seed 0 of the present and of the carton and on seeds 0 and 1 of the lid box are finite, without intersections and at rest afterwards as well (hands at 2.5-3.1 m/s, 403x on the present). Finger links are pressed into the items with up to 253 N in the rows above too; that is the squeeze §6.3 cites.

11 Runtime

Scope: wall-clock seconds per simulated second, against the 200x limit. The rows of the first table use two solver threads unless a row says otherwise, the robot model on the CPU. ***They were measured on a starved machine:*** *the host's 16-CPU quota was shared with other jobs and exhausted in 95-100 % of scheduler periods during the runs (each record carries the share,* `code.threads.cpu_quota`*). A row says what a run cost there, not what the code costs on a free machine; the Newton iterations per step beside it do not depend on the machine.*

Verdict: free moves and rest are inside the limit; anything in which paper is pushed, dragged, gripped or opened is above it.

What runs	wall/sim	Newton iterations per step; steps at the cap
GAR `act()`, free-air actions with an RGB observation after each: home / unwrap-gift	53x (physics 46x)	–
GAR `act()` on counter / open-bag (tote with handles on the stack), free-air actions: physics only (the run was stopped after three actions; with the shared GPU full each RGB frame took 446-648 s, so no total with RGB was measured)	81x (two actions after the first, which warmed up at 280x)	–
Rest: lid box / wrapped present (30 s; 10 s) / tote on the stack (60 s) / carton	30x / 193x; 107x / 127x / 54x	1.00-1.06; 0-3
Opened tote standing (10 s; 3 s); energy audit	40x; 35x; 209x	1.3; 1.1; 7.3 (73 of 525)
Lid air film, phantom lift	30-32x	1.7-1.9; 0

What runs	wall/sim	Newton iterations per step; steps at the cap
Robot lid removal, scripted: 10 cm/s (five runs); 1 cm/s on one thread	40-52x; 69x	1.4-1.5; 0-3 of 3,375
Robot lifts an opened tote by its rim	59x	1.9; 5 of 2,900
Legal abuse: lid box (three seeds) / carton	19-32x / 33x	4-39 of 2,200 / 23 of 2,200
Legal abuse: wrapped present (two seeds)	**242-289x**	119-223 of 2,200
Lifting the tote by a handle, then pulling the handle with 40 N	**392x**	7.2; 86 of 2,075
Legal abuse on the counter, the hands reaching the tote (refused commands re-issued, four threads)	175x	193 of 2,231
Assisted carton opening (tape peel, four flaps), three runs; the same with film	115-117x; 198x	5.4; 95-100 of 3,862
Robot arm pulling the present's taped end (assisted grip)	**244x**	5.3; 99 of 2,025
Assisted unwrap attempts, nine runs on two or three threads	86-**609x**	6.3-6.9; 157-230 of 2,900-4,100
Hand-held tote opening, four threads: lift, flap, parting the bands / hand goes in / set-down	**880x / 590x / 460x**	steps at the cap: 120 of 1,800 / 271 of 1,100 / 270 of 1,075

A render-worker start takes 10-11 s. The worker is kept 15 min after the last image request (§12), so a 0.02 s command after a minute without an image request costs 146x.

Paper contact. The Newton matrix is assembled straight into a persistent sorted pattern, the PARDISO analysis is reused while the pattern holds, the crease return mapping is batched, and one broad phase serves each step. On the contact phase of the wrapped present this takes a run from 1,157x to 521x (0.555 → 0.237 s per Newton iteration; free moves 139x → 72x) without changing any energy term, tolerance or iteration rule: from the same state, 250 contact steps differ by at most $1.3{\cdot}10^{-10}$ m with the same Newton count in every step. With the robot pressing, dragging and gripping the wrap through GAR with RGB the simulation runs at 481x. **It is above the limit.** What remains is the number of Newton iterations: 8 per step on average in contact and 24, the cap, in about one step in forty; in those steps the collision check allows the full step and the energy line search takes a fraction of it. Keeping contact-free robot links out of the Newton system and a coarser mesh for buried layers were not tried.

Correct friction costs iterations; one remedy was measured and is not adopted. With the stopping rule of §3.2 a sliding contact takes 4-10 Newton iterations per step (a rule on the update size alone stops after one, before the contact has moved). Every step starts with each sliding contact at zero tangential displacement, in the stiff stick branch of the friction law, and the first iterations find out again that it slides. Starting the iteration from the last step's motion instead (when that move is collision-free) was tried as a prototype outside the world: a sliding block went from 4.4-7.2 to 1.1-2.1 iterations per step at the same effective friction, and the carton's tape peel from 250x to 197x with the same pinch force, bonded area and seam fraction at 10 of 11 logged times. On the wrapped present it did nothing where a hand pushes the paper (12.8 iterations per step, 64 of 125 steps at the cap, with it and without) and cost a third more iterations in the first half second, while the items settle and the guess sends them on. It is not in the world: the gain is not where the cost is.

Where the iterations go (measured outside the world; none of it is in the world):

- *The stopping rule itself is cheap.* On the tote it costs nothing; on the wrapped present about 6 % more iterations. It waits only where paper pairs are being unloaded toward the edge of the barrier's range.
- *The steps that reach the cap are friction kinks.* While the tote is lifted, 19.5 iterations per step on average, the largest update sits on the back rope handle: rope and paper contacts switching between sticking and sliding inside one step. A friction re-solve with a predictor, kept as an opt-in module outside the world file, brought that phase to 10.4 iterations per step, the counter drags down 39 % in wall time and the wrapped present 18 %. It is not wired in: its converged states differ from the unmodified solver's by 20-30 µm (median), more than the 10 µm tolerance, and it was not carried through the benches.
- *A phantom hand at its force cap crawls.* A force-capped driven body advances by (net force)/(spring stiffness) per iteration, 12 µm for a 10 N hand: 17 iterations for 0.2 mm, the cap of 24 for anything faster, and the iteration can end with the hand short of where its force would take it. This is most of the iteration count of the assisted demonstrations above. A step-doubling line search makes the small case worse.

Thread budget: on the wrapped present at rest, 0.36 s of linear solves per contact step with no thread cap against 0.12-0.14 s at 2-8 threads; the environment caps at `GIFT_BAGS_THREADS` (default 4).

12 Interfaces

Scope: how a policy or operator reaches the environment.

- `gar list`, `gar inspect gift-bags`, `gar tasks inspect gift-bags`, `gar tasks resolve`, `gar run`, `gar action`, `gar observe`, `gar state`, `gar verify`, `gar dashboard`, `gar export --include-traces`, `gar pause`, `gar stop`.
- **The six tasks are named where a user looks for them.** `gar tasks inspect gift-bags` prints the presets (`counter`, `home`, `receiving`) and the task names (`open-bag`, `bag-purchase`, `lift-lid`, `unwrap-gift`, `open-present`, `open-carton`), and `gar run gift-bags --task NAME` starts that task: the preset follows from the task.
- Policy observations: `overhead_rgb` (ceiling drop mount over the work surface) and the two D405-envelope wrist cameras; joint positions and velocities; the task instruction. Diagnostic renders (`room`, `front`, `counter_wide`, `bag`, `mat`, `gift`, `side`, `carton`) are privileged. The policy key gets health, capabilities, task, observation and action; state, evaluation, history, variations, geometry and checkpoints return 403.
- Actions: named joint targets with a tick count (a timed motion, below); Cartesian pose/delta; Franka Hand half-gap (position) or total inward effort (force mode, $\leq$ 70 N). One tick = 4 ms. Assisted fixtures are not reachable from the backend (`GiftBagsEnv(assisted_scene_hook=...)` is for experiments only).
- **Refused commands do not break the session.** A command is planned and validated in full before anything is written to the robot: duration, arm, IK path, joint names and limits, grip force and width. A refusal raises `ActionRejectedError`, or `UnsupportedActionError` for an unsupported kind, and leaves the session usable. Over HTTP, five malformed commands each return 400; health stays `runtime_ok`, and the next legal command returns 200 without owner recovery.
- **Joint-target commands are timed motions too, and refused when too fast.** Every commanded arm joint moves from where it is commanded to be to its target over the command's ticks on a minimum-jerk profile, and the command is refused, with the minimum duration in the message, if that asks a joint for more than its velocity limit ("right/fr3_joint1 would need 91.28 rad/s (limit 2.62): give this motion at least 0.70 s (175 ticks)"). The same 1 rad move over 0.72 s runs with no velocity reflex. Finger joints are set at once.

  A target set at once would be a step for the position servo, which drives the joints far past their velocity limits; the reflex only freezes the targets. In the abuse probe (lift-lid, seed 2) such step targets make the right arm trip the reflex 65 times in 0.5 s, its tool point move 97 mm in one 4 ms step and joint 4 end 0.15 rad beyond its limit, after which the next command is refused because the joint already stands outside its limits. A real FR3 rejects such a command before it moves. The joint limits of the robot simulator are soft, and a hit at that speed passes them; no legal command reaches that.
- **Cartesian commands are timed straight-line motions, as a motion generator executes them.**
  - The TCP moves on a straight line from where the arm is commanded to be, with orientation slerped, over the command's duration, on a minimum-jerk time profile.
  - The path is solved before anything moves: one IK branch, poses at most 1 cm / 8° apart.
  - It is refused if unreachable, if it needs a branch change, or if the duration asks a joint for more than its velocity limit. The message says the minimum duration.
- **Motion-safety guards are honoured.** `stop_on_contact`, `stop_on_robot_contact` and `stop_on_arm_arm_contact` (default on): a guarded command stops after the contact step in which the guard fires and reports `stopped_on_contact` and `stop_reason`. Contacts come from two places:
  - penetrating MuJoCo contacts between different bodies (robot against furniture, floor, itself or the other arm; a wrist camera's bolted mount on its own arm is excluded);
  - contact-world reactions on robot links (paper, tape, items).

  Measured through the GAR action path on home/lift-lid:
  - in every configuration (dual-fr3, dual-panda, single-fr3, single-panda), a guarded descent onto the gift-box lid stops on the fingertip's contact with the lid in the contact world: the TCP 9.5 / 11.2 mm (FR3 left / right) and 8.8 / 9.9 mm (Panda left / right) above the lid top, with 0.4–21 N on the touching finger; the hand then closes and opens and the arm moves back up, with no velocity reflex; after each arm the base has

not been lifted and the frame is still in it;

- with the other hand still over the box, the arm-arm guard stops the second arm 104 mm up, so each arm returns to ready before the next.

- **Per-step physics trace** of every HTTP command. Each 4 ms step records:
  - joint positions, velocities, targets and servo targets;
  - hand commands and item poses;
  - Newton iterations and convergence;
  - reactions on robot links;
  - sheet vertices on the first step, every 25 steps and the last.

  Bounded at 64 MiB / 100,000 items per session, oldest first.
- **One render worker per session.** Reused between operations; released after 15 min without an image request and at close; replaced and retried once if it dies. With the worker reused, a 0.02 s free-air command takes 1.00-1.04 s end to end; with a new worker 10.35-10.94 s (home/lift-lid, overhead RGB). The idle release is 900 s because a policy that thinks for more than a minute between actions would otherwise pay a full worker start (10.0-11.0 s, 501-548x for a 0.02 s command) on every action. `GIFT_BAGS_RENDER_IDLE_S` sets it for `gar run` sessions (which have no renderer options), `renderer_options idle_release_s` for Python callers. After a release the first image still costs the 10-11 s start.
- **A refusal tells the policy why.** The HTTP gateway replaces a backend's error text with a fixed sentence for restricted callers. For planning refusals, which are computed only from the command and the robot's own joints (duration, joint jump, IK branch, limits, grip width), `gift_bags` attaches the reason and the gateway appends it ("... Reason: joint jump 0.40 rad (IK branch change)"). Other backends do not attach a reason: without the attribute the fixed sentence stands alone (`http_server._error`, `policy_reason`).
- **Variations.** `GET /v1/variations` returns the catalogue (presets, tasks, seeds, pose ranges). A session's variation is fixed at launch; other variations are refused cleanly.
- **GPU memory, and running without the card.** A session keeps the robot model's arrays on the GPU (about 0.7 GB; the robot solver itself is MuJoCo on the CPU and the contact world never uses the GPU), and a render worker for RGB observations takes about 4 GB. When other jobs fill the card, a new session dies at start with a CUDA out-of-memory error. `GIFT_BAGS_DEVICE=cpu` (or `device="cpu"` in Python) puts the robot model on the CPU: physics, state and evaluation then need no GPU (home/lift-lid stepped at 34x on one thread that way); images still do. The doctor reports the free memory against both needs, and every record stamps the device it ran on. A film whose frame fails to render is retried in a new worker for up to twenty minutes per frame instead of being lost. The experiments in this report were run with the robot model on the CPU; the two devices have not been compared number for number.
- **Doctor.** Checks the thread setting against the container's CPU quota (cgroup `cpu.max`) and the runtime dependencies (`trimesh`, `usd-core`, `threadpoolctl`). Checkpoints say they are in-memory for the session.
- **Robot setups.**
  - `gift_bags/robot_setup.py` validates setups by content. The catalogue's `label-fr3` rig (gift-fr3 under another id) and gar.robot-setup.v1 objects with qualified content are accepted. Other setups are refused with the physical reason. A hand-built setup the catalogue cannot resolve (for example another gripper with a mount profile it does not list) gets this environment's reason first ("one franka-hand@1 on left/flange ...: the paper contact world couples a two-finger parallel gripper with fingertip pads along one closing axis") with the catalogue's message appended.
  - The scene's FR3-named ready pose applies by joint number to any Franka arm and fails on unknown joints, so the dual-Panda pair starts in the same pose as the FR3 pair.
  - Every configuration is exercised through the action path (above).
- **A preset that cannot be reset is reported, not crashed into.** `variants.availability()` reads each preset's start-state cache for the current code: a cached state that is unsound (the tote torn while it was made) makes the preset unavailable with that reason in `GET /v1/variations` (`availability`, `unavailable_tasks`, `default`), in `gar tasks resolve` and at backend start (a refusal, not a RuntimeError), and `gar doctor gift-bags` lists the start state of every preset. A start state that is merely not built yet is available: the first

launch builds and caches it.

- **Start states.** Each start state is named by its key. A world looks for it in `gift_bags/start_states/` (shipped with the code) first and in the local cache (`.cache/gift_bags/start_states/`) second, and builds only what neither has. The code ships the tote's shipping state, the tote with its handles, the sealed carton and two of the 18 wrapped presents (three boxes x three lid clearances x two wrap stocks); the other wrapped variants build on first use. `python -m gift_bags.prebuild` prints the table (key, shipped / cache / missing) without building; `--build` builds what is missing, `--ship` copies the current keys into the shipped folder and removes files no current key names. `gar doctor gift-bags` and `GET /v1/variations` (`availability.<preset>.start_states`, `start_states_ready`) give the same table per variant. An edit to a generator changes its keys, so its states are built again. Build times from nothing (the build logs and each file's `info`): tote shipping state 24 and 30 min (two builds, 4 threads, load average 15-25), handle assembly 8 min, each wrapped present 10-17 min (about four hours for all 18 one after another), the carton 23 s.
- **The G1 humanoid is not a robot choice.** The rig `gift-g1` and `gift_bags/g1.py` exist (the floating G1 with Dex3-1 hands stands hands-off on its PD servos at the counter and the dining table for 10 s: pelvis drift 0.17 mm, tilt $\leq 0.39°$), but it has no balance controller. Moved from the ready pose toward the gift-box lid (three independent runs), its hands press on the table top, the pelvis is pushed back, and it lies on the floor within 2–4 s (90° of tilt); its hands start below the table top, a 5 mm downward hand move is refused by the path check, and no Dex3 hand has touched a packaging item. `g1` (and the same hardware under the ids `gift-g1` and `scan-g1`) is therefore refused with that reason; `GIFT_BAGS_G1_DEV=1` lets development code build it. What it needs to be offered: a balance or whole-body controller under arm motion, a stance from which the hands reach the work surface, and one task interaction through the action path.

13 Limitations

Read before relying on the environment. These are gaps between the real activity and this simulation, not disclaimers.

1. **Opening the flat tote with the holds of two hands is not shown** (§8.2). One hand lifts the bag, the other flicks the flap down, both part the bands: the mouth opens fully and nothing tears at the gusset feet. The hand inside then fails in every form tried: turned flat at the faces' bottom edges it is above the bottom, which hangs 45-50 mm lower as a closed V; pressed on the mat it bridges the V; sent into the V it is squeezed out again at 10 N. The bottom's two ends stay folded in the gusset feet throughout. The openings that do reach the goal state (§8.2, mould) guide 2,760 of the tote's 3,559 vertices and tear the bag. Until a hand-like sequence squares the bottom, `open-bag` and `bag-purchase` are not shown to be completable. No robot has opened the tote.
2. **A free flat tote's folded flap rests 16 mm up, not flat** (§6.4). The shipping state is sound (no tear, the fold pressed and aged a week), but when nothing presses on the bag the flap opens about 21°. In a stack or a carton a real tote lies flat; how far a single free one opens was not measured on a real bag.
3. **Rope handles: no robot has used them.** The rope is threaded through the hole with its end in the knot, and the attachment is tested by lifting the tote by a handle (§6.4). The grasp a clerk uses most, by the rope loops, has not been demonstrated by a robot. The rope holes are one mesh cell, 10 mm squares; a real hole is about 5 mm round. A hole that small needs a finer mesh around it and room for the rope's own contact range, and was not built.
4. **Tape.**
   - Office tape on gift wrap: a 90° peel runs at 3.44 N in steady state against 2.85 N from the energy balance (1.21x) with spikes to 11.8 N; the tape mesh (2.5 mm) is coarser than the process zone (0.6 mm) (§6.9).
   - Packing tape on the carton: a 90° peel is within 2.3 % of 9.6 N at 2, 4 and 8 mm cells. The 9.6 N is the ledger's bond energy on a kraft liner (200 J/m^2, an estimate) times the tape's width, not a data-sheet value: the data sheet gives adhesion to steel only, and no peel value on fibreboard is published. **A 180° peel is wrong: 18.7 N against 4.8 N** (§6.11). A policy that folds the tape back on itself pays about four times the real force.
   - The wrapped present's generator leaves 2.3 N net on the most loaded end tape; in the task world the end tapes hold 6-7.5 N at rest (item 22; §6.5, §6.9). What a real end tape holds was not measured.

5. **Tearing.** The out-of-plane tear force of the wrap is 0.52 N against a target of about 0.45 N at the scene's 8 mm mesh, and within 3 % between 4 and 8 mm cells; below 4 mm it is not qualified (1.0 N at 2.5 mm with yielding legs). Where a tear starts is an estimate from the stock's strength and toughness, not a measurement (§6.6). The wrap is one stock per seed; printed-foil and tissue wraps are not modelled.
6. **Receiving is shown only with assistance, and only by peeling.** Phantom hands peel the tape from its courtesy tab and fold the flaps out; no robot has done it. Slitting the tape with a knife, the usual act in a stock room, is not modelled, and the instruction says so (there is no knife; peel from the tab). The evaluator counts a flap propped on another flap as open in the score (§6.11). The kraft shopping-bag variant from the research design is not built.
7. **Robot coupling is explicit with a one-step lag.** Pushes with link surfaces or outer finger faces reach the robot one contact step late (filtered, capped, extrapolated inside the substeps). A bimanual squeeze on the rigid lid holds a steady ∼20 N after a 0-40 N transient lasting ∼0.6 s; stiffer squeezes or lighter links may still chatter. Grasps between the finger pads are resolved inside the contact world.
8. **Runtime with paper in contact is far above the 200x limit, and many such steps end unconverged** (§11). Free moves and rest are inside the limit. Coulomb friction (§3.2) costs iterations; a step that reaches the cap of 24 iterations is accepted with its contact forces not settled.
9. **Flat-stack approximation.** Only the top tote is a sheet with handles; the four below are rigid plates with the flat outline and no handles, so the stack is 10 mm high where five real totes with their handle lumps would be several centimetres.
10. **Single contact caliper per world.** Tape and secondary sheets use the main stock's caliper (§3.1).
11. **Uncalibrated estimates** carry [E] in §5: paper through-thickness compressibility (5 % at 1 MPa), the tote's crease law (board data scaled by t^2; §6.10 gives the paper-class bracket), the press that sets the shipping fold (200 kPa), the storage pressure and times, friction coefficients (every pair), tape-on-paper toughness, finger and link compliance; how far crumpled paper can be crushed (to half its length) and the tenth of the yield strain beyond which it crumples again (§6.4), the 1 mm of hand slack in the wrap's girth (§6.5), and the fraction of the transition velocity below which a contact keeps its stick displacement (§3.2).
12. **No humanoid.** The Unitree G1 only stands here: it has no balance controller and falls when it reaches, and no Dex3 hand has touched a packaging item. It is refused as a robot choice with that reason (§12). What it needs is listed there; none of it is built.
13. **Whether an opened but unworked tote passes the gusset checks is not established.** With the shipped (board) crease law an opened tote holds its gussets at 140°; with the paper-class law, which is as defensible for this stock, they fall back to 93°, under the 120° limit (§6.10). Both ends of that bracket are soft (the paper law's anchor is the smallest angle reached at saturating press force, and its time rate has no unit in the source), and the limits themselves (120°, mouth 0.80) are authored, not measured on real bags. The catalogue says so.
14. **Variation is narrow.** The home preset varies the box, the lid's clearance, the wrap stock and the pose; the counter varies pose only (each tote size or stock needs its own hour-long shipping state, not built); the receiving preset varies pose only.
15. **No autonomous or policy solution exists**; scripted robot demonstrations are component checks (§9).
16. **Friction is Coulomb's law with estimated coefficients, converged to a tolerance.** A contact below its limit holds (with an elastic offset of at most 4 µm) and one above it slides with the ledger's coefficient (§3.2). The iteration settles contact forces to 5 % of the largest contact force in the world, so a light contact beside a hard squeeze is settled less tightly than it would be alone, and a step that ends at the iteration cap is not settled at all. No friction coefficient here was measured on these materials.
17. **Props are fixed boxes.** The till, the wrapping supplies, the cabinet, the shelving, the chairs, the table legs and the other presents stop an arm or a pushed item but cannot be moved, tipped or reached into (§2.2).
18. **The arm's inverse kinematics does not check self-collision.** A commanded pose can have a solution in which the wrist runs into the arm's own forearm; the robot simulator then resolves the contact physically and the hand ends off its target (in the lid demonstration this skews the right hand by 17-22° unless the hand is turned half a turn, §9.3). The guard `stop_on_robot_contact` stops such a command; a scripted demonstration has to read the reached pose.
19. **The same commands give the same run only on one solver thread** (§9.3). On several threads a result

near a limit can differ from run to run.

20. **Sessions and films depend on a shared GPU unless told otherwise** (§12): 0.7 GB per session for the robot model unless `GIFT_BAGS_DEVICE=cpu`, 4 GB per render worker. The experiments in this report were run with the robot model on the CPU; the two devices were not compared number for number.
21. **Unwrapping the present is not shown, even with assistance.** Phantom hands peel the end tape (7 N) and open the end; nine attempts to get the paper off then failed (§8.3). By sliding: the paper travels with the box unless it is held firmly, and held firmly (40 N) the box jams after 6 mm against a side flap that has swung back across a corner of the opening. By tearing: the paper tears at hand forces, and the one sequence tried bared a tenth of the box. What was not tried: a hand that keeps the flaps open while another holds the paper and a third draws the box (three hands); tearing the top open from end to end and lifting the box out; turning the present over and peeling the seam. No robot has unwrapped the present or opened one taped end (§9.4).
22. **The present is wrapped empty and filled afterwards.** Its start-state generator wraps the box without the 350 g photo frame; in the task world the box then sinks 0.3 mm into the loose paper under it and the end tapes hold 6-7.5 N at rest instead of 2.3 N (§6.9, proven by lifting the frame: the load goes and comes back). No bond softens and the present rests. The generator should wrap the box with its contents; that changes all 18 wrapped-present start states and every record on the present, and is not done.
23. **The joint-target rule's least duration has no margin.** A named joint target that is too fast is refused with the least duration it would need (§12). Sent again with exactly that duration, the motion runs at the joint's velocity limit and trips the robot's velocity reflex: 179 reflexes in 8.6 s of abuse on the counter (§10). The world survives it; a client following the message does not get a clean motion. The least duration should carry a margin, and the abuse probe should then be repeated on every preset. Not done.

14 Reproduction

The environment ships as the `gift_bags` package of RoboWorld. The experiments of this report were run by scripts that drive `GiftBagsEnv` and the contact world directly; those scripts are not part of the release. What the release runs:

What	Command
Discovery / dependencies	`gar inspect gift-bags`; `gar doctor gift-bags` (also reports each preset's start states); `gar tasks inspect gift-bags` (presets and task names)
Managed session	`gar run gift-bags --task lift-lid`; or `gar tasks resolve gift-bags --select preset=home --select task=lift-lid --output t.json`; `gar run --task-file t.json`
Start states: table, build, ship	`python -m gift_bags.prebuild`; `... --build [--only PRESET] [--seeds 0,1,...]`; `... --ship`
Visual assets	`python -m gift_bags.visual_assets` (fetches the models and textures and checks their SHA-256 pins)

Settings: `GIFT_BAGS_THREADS` (solver threads, default 4; the doctor checks it against the CPU quota), `GIFT_BAGS_DEVICE=cpu` (the robot model on the CPU, §12) and `GIFT_BAGS_RENDER_IDLE_S` (the render worker's idle release). The same commands give the same run only on one solver thread: on several threads the parallel linear solves are not bit-reproducible (§9.3).

Start states (the tote's shipping state, its handles, the wrapped presents, the carton) are files named by a key made from the generator's source and its physical inputs; a changed key means a rebuild (§12). The films are rendered from recorded states only (no re-simulation, no splices).

Further figures

Further figures drawn from the recorded experiments of the sections named in their captions.

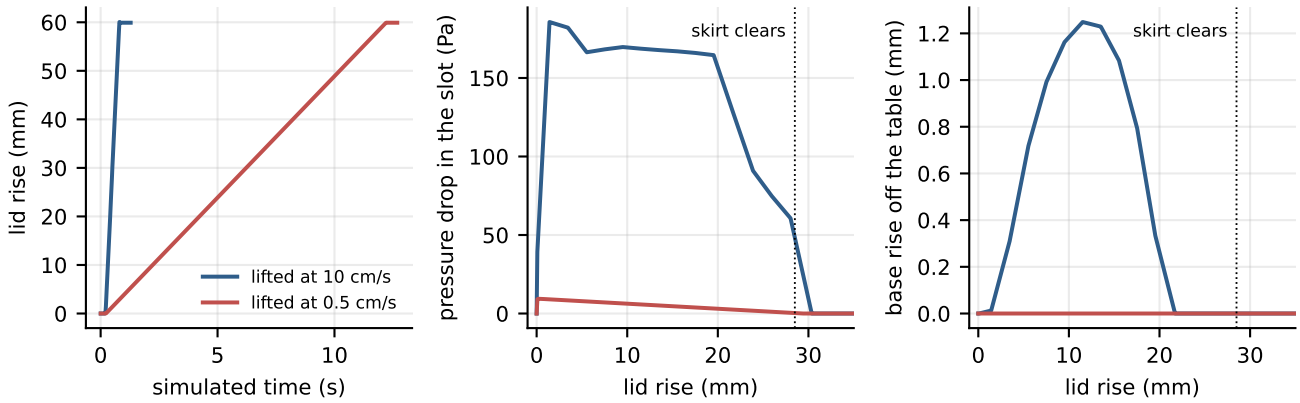


Figure 18. Lid air film under the ASSISTED phantom lift of §8.1 at two speeds. Left: lid rise over time. Middle and right: the pressure drop in the slot and the base's rise off the table against the lid's rise; the dotted line marks the rise at which the skirt clears the base.

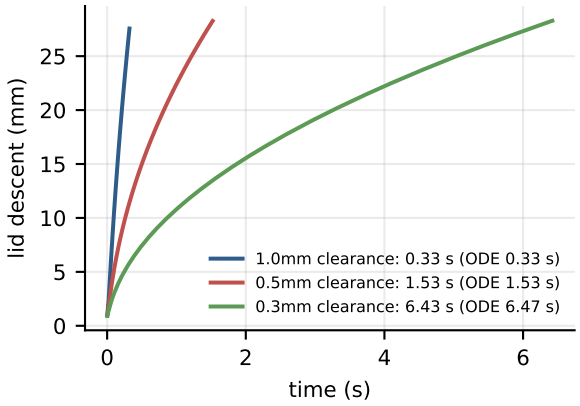


Figure 19. Lid descent under its own weight at three skirt clearances (§6.2): the lid's path in the coupled implicit step, with the descent time of the simulation and of the scalar ODE of the same film law in the legend.

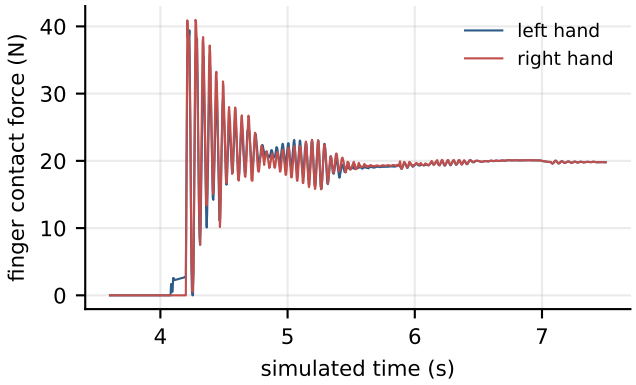


Figure 20. Largest finger contact force of each hand during a bimanual squeeze on the lid and a 2 cm lift, every 4 ms step of the squeeze bench (§9.3).

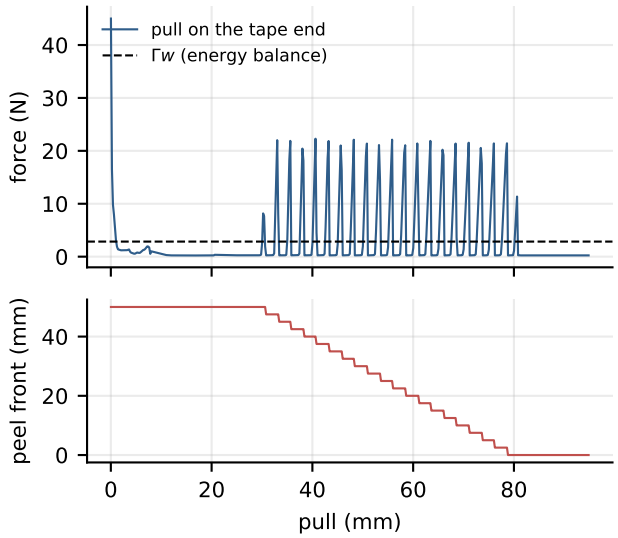


Figure 21. 90° peel of a 19 mm tape strip (§6.9, one bond per tape vertex). Top: pull force on the tape end, with the energy-balance force Γw dashed. Bottom: position of the peel front. Both against the pull.

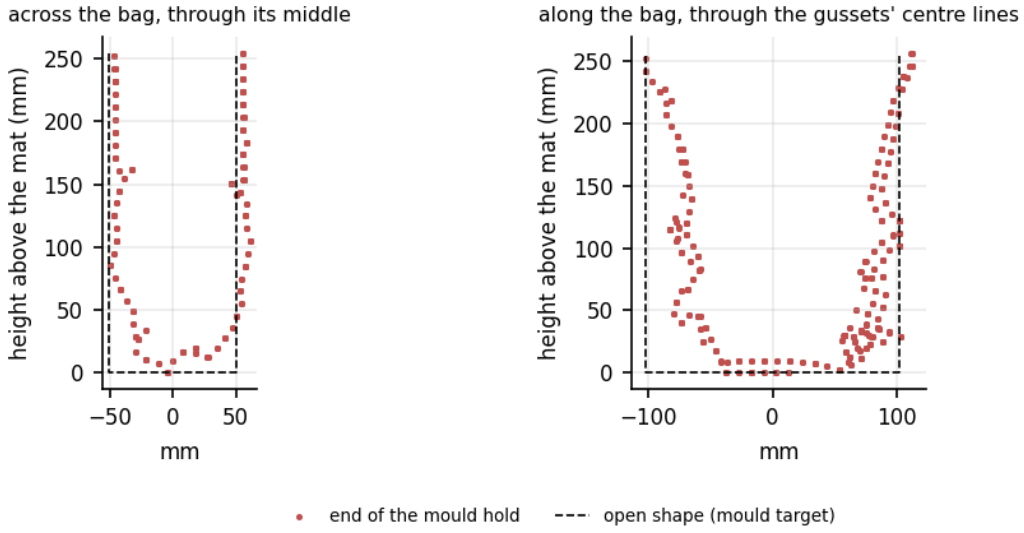


Figure 22. The tote at the end of the 2 kN/m mould hold of §8.2, sliced through its middle: the points are the recorded outer paper surface within 6 mm of each plane, the dashed line is the open shape the mould pulls toward.

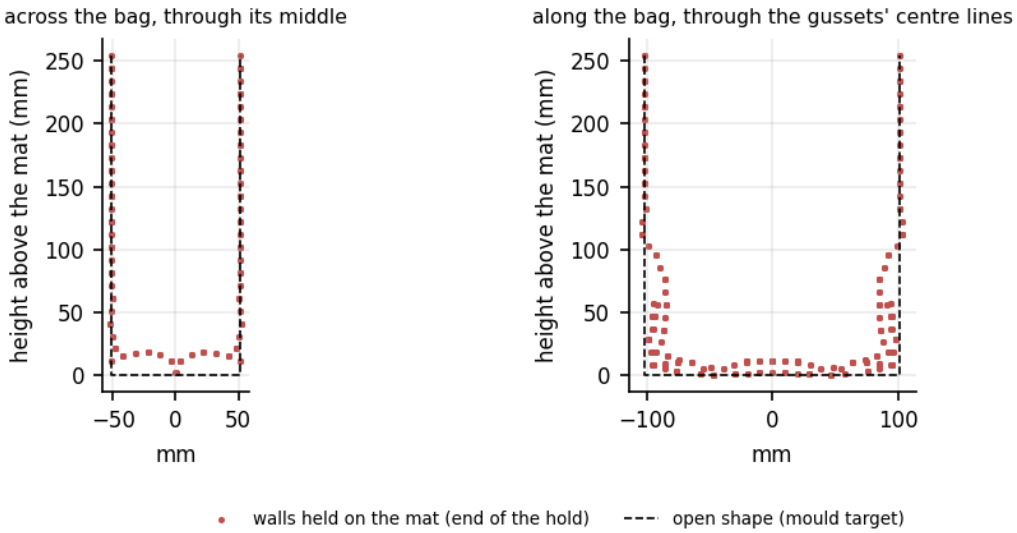


Figure 23. The tote held open on the mat along its folding motion (§8.2, walls held), at the end of the hold, sliced as in the mould figure.

Films and views

Each strip shows frames of the film of an experiment at the playback times given, with the film's own caption bar. Each still is a render of a start state. A film shows the scene as it was when its experiment was run.

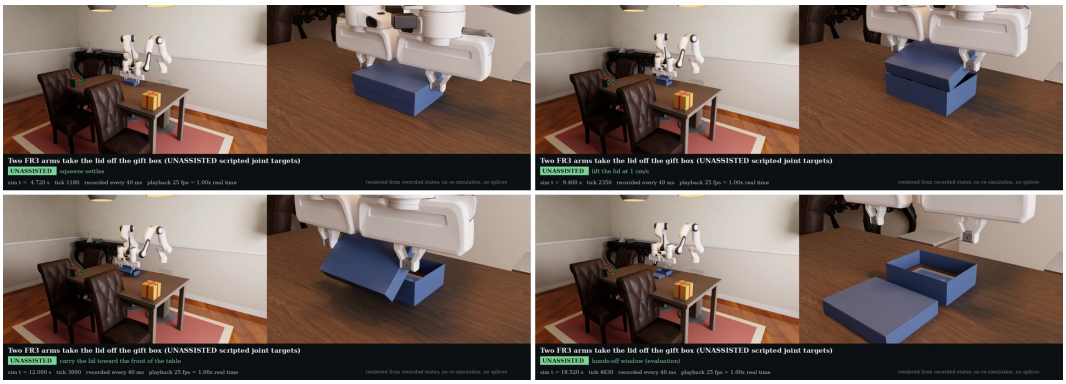


Figure 24. Two FR3 arms take the lid off the gift box (UNASSISTED scripted joint targets). Frames at 4.7, 9.4, 12 and 18.5 s of playback, caption bar included. The 1 cm/s removal of §9.3.

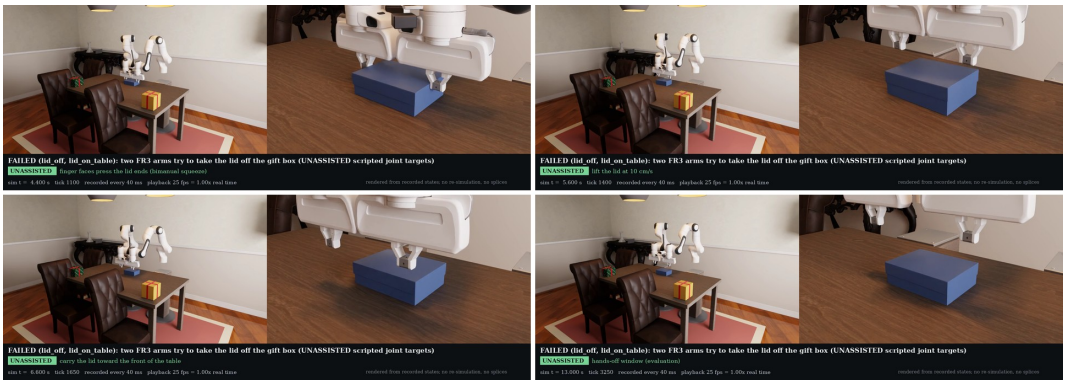


Figure 25. FAILED (lid_off, lid_on_table): two FR3 arms try to take the lid off the gift box (UNASSISTED scripted joint targets). Frames at 4.4, 5.6, 6.6 and 13 s of playback, caption bar included. The 10 cm/s removal of §9.3.

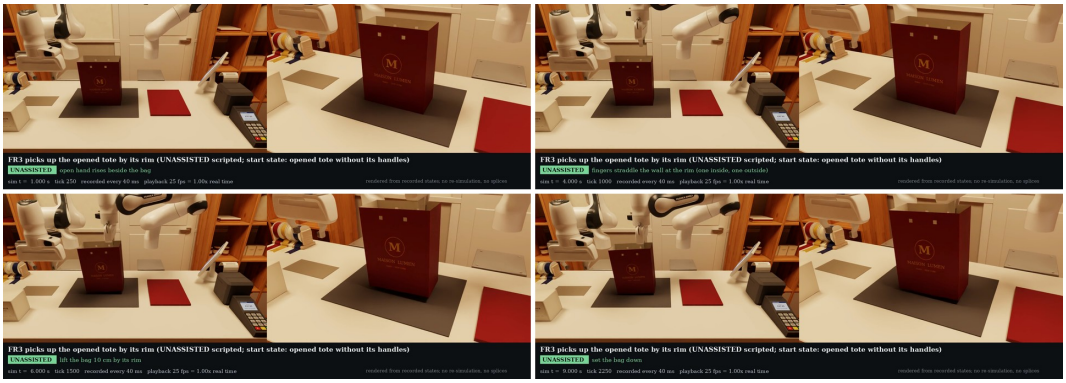


Figure 26. FR3 picks up the opened tote by its rim (UNASSISTED scripted; start state: opened tote without its handles). Frames at 1, 4, 6 and 9 s of playback, caption bar included. §9.1.

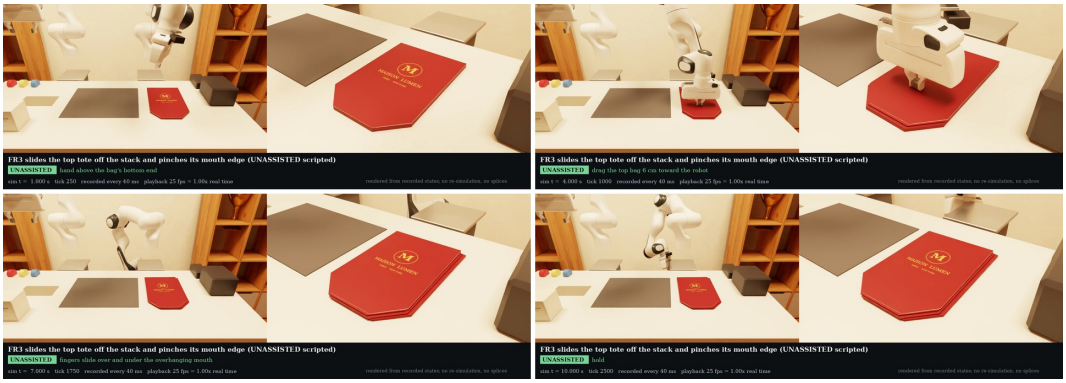


Figure 27. FR3 slides the top tote off the stack and pinches its mouth edge (UNASSISTED scripted). Frames at 1, 4, 7 and 10 s of playback, caption bar included. The failed second run of §9.2 (IK branch flip).

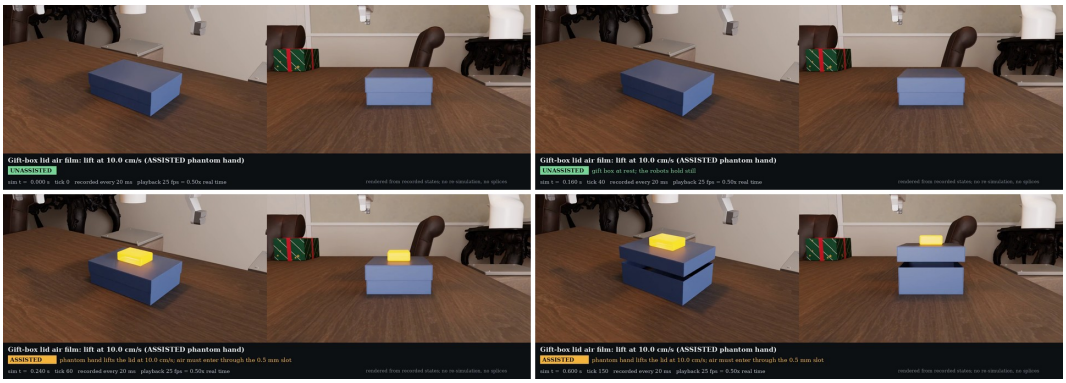


Figure 28. Gift-box lid air film: lift at 10.0 cm/s (ASSISTED phantom hand). Frames at 0, 0.3, 0.45 and 1.2 s of playback, caption bar included. §8.1.

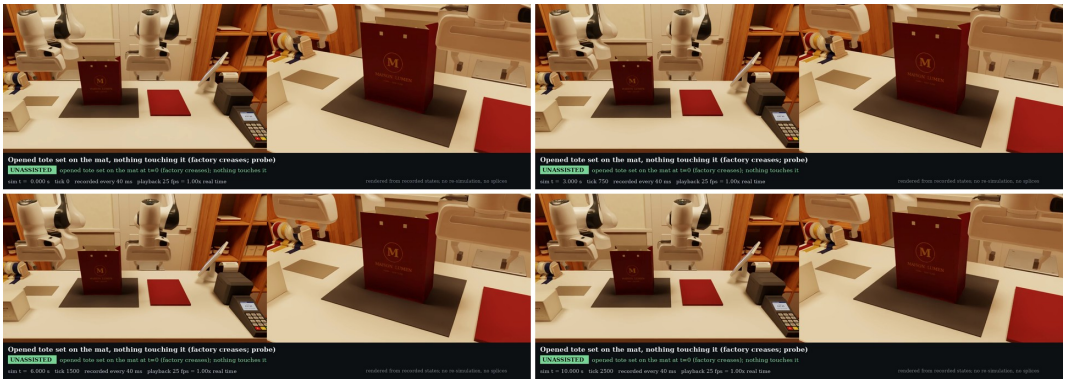


Figure 29. Opened tote set on the mat, nothing touching it (factory creases; probe). Frames at 0, 3, 6 and 10 s of playback, caption bar included. §6.7.

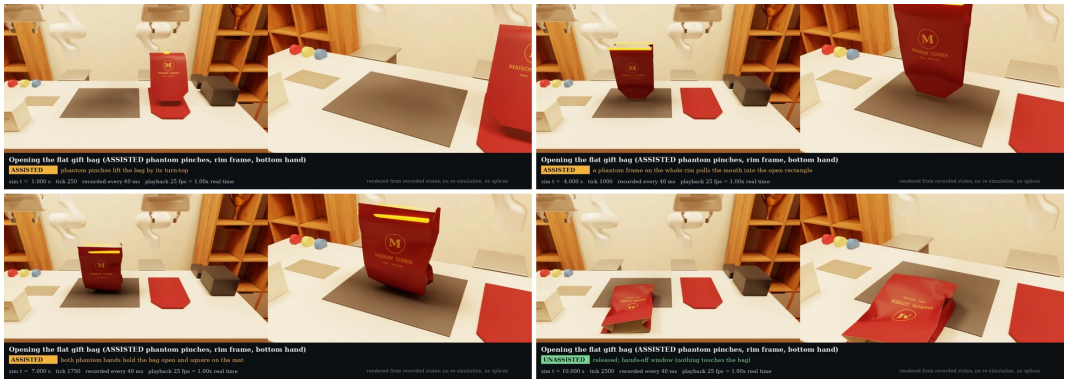


Figure 30. Opening the flat gift bag (ASSISTED phantom pinches, rim frame, bottom hand). Frames at 1, 4, 7 and 10 s of playback, caption bar included. Attempt 6 of §8.2, a failed run.

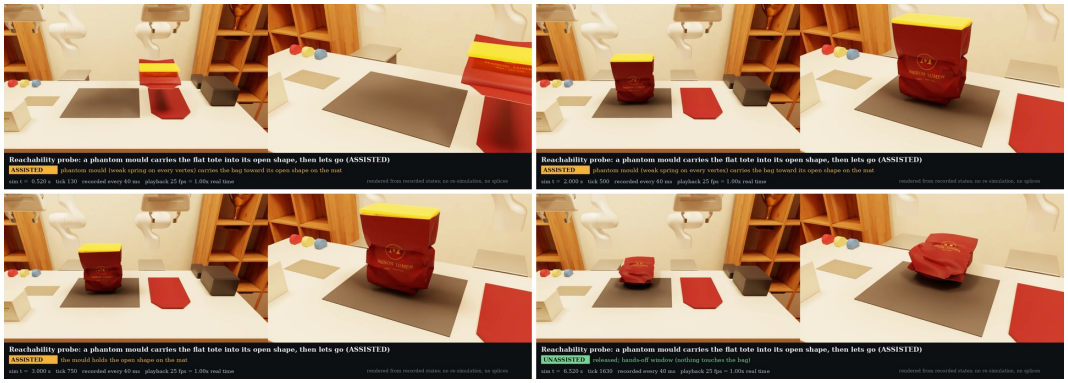


Figure 31. Reachability probe: a phantom mould carries the flat tote into its open shape, then lets go (ASSISTED). Frames at 0.5, 2, 3 and 6.5 s of playback, caption bar included. The 2 kN/m, 0.5 N cap mould of §8.2.



Figure 32. Opening the flat tote along its folding motion (ASSISTED phantom hands on the walls; bottom free), then letting go. Frames at 2.2, 5, 6.6 and 8.8 s of playback, caption bar included. §8.2, faces and gussets held above their diagonals.

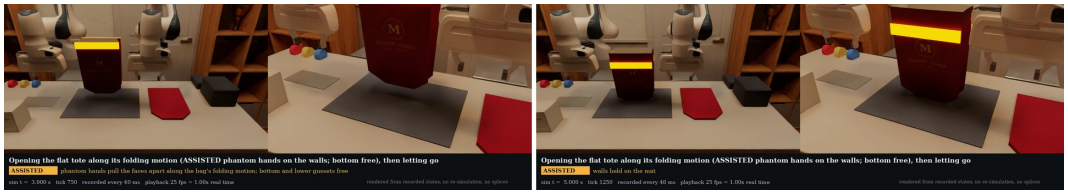


Figure 33. Opening the flat tote along its folding motion (ASSISTED phantom hands on the walls; bottom free), then letting go. Frames at 3 and 5 s of playback, caption bar included. §8.2, only the faces' turn-top bands held.

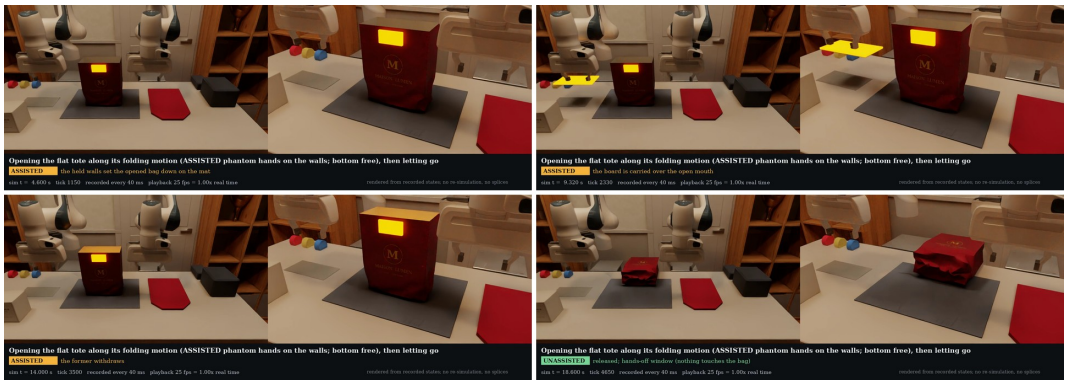


Figure 34. Opening the flat tote along its folding motion (ASSISTED phantom hands on the walls; bottom free), then letting go. Frames at 4.6, 9.3, 14 and 18.6 s of playback, caption bar included. §8.2, walls held, then former and board.

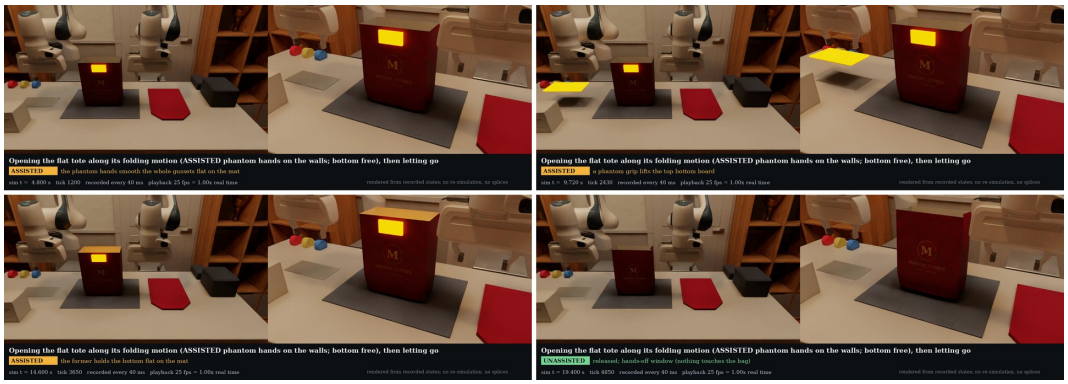


Figure 35. Opening the flat tote along its folding motion (ASSISTED phantom hands on the walls; bottom free), then letting go. Frames at 4.8, 9.7, 14.6 and 19.4 s of playback, caption bar included. §8.2, DIAGNOSTIC run: the bottom-perimeter creases given the open shape's 90° as plastic memory, gussets held flat, with former and board.

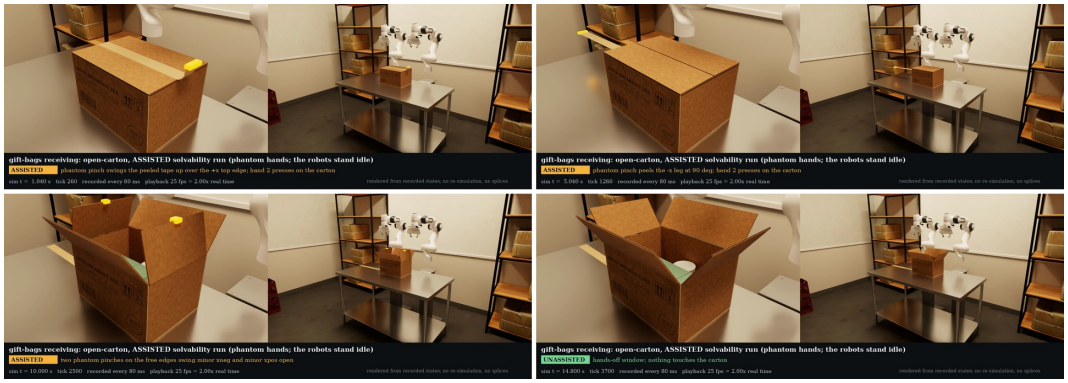


Figure 36. gift-bags receiving: open-carton, ASSISTED solvability run (phantom hands; the robots stand idle). Frames at 0.5, 2.5, 5 and 7.4 s of playback, caption bar included. The assisted opening of §6.11.



Figure 37. The counter preset: the boutique cash-wrap counter.

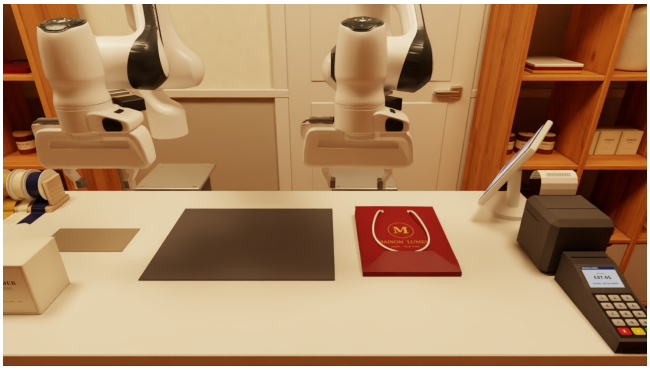


Figure 38. The counter preset: the stack of flat totes on the counter mat.



Figure 39. The counter preset: overhead camera.



Figure 40. The counter preset: left wrist camera.



Figure 41. The counter preset: right wrist camera.



Figure 42. The home preset: the dining table with the gift box.

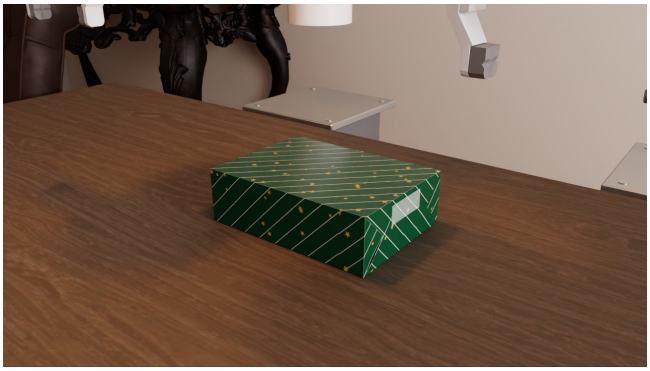


Figure 43. The home preset: the wrapped present.



Figure 44. The receiving preset: the receiving table.



Figure 45. The receiving preset: the taped carton.



Figure 46. The receiving preset: overhead camera.

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